

Tracking maternal proteins uncovers a central role for the residual body in organelle recycling during *Toxoplasma gondii* replication

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eLife Assessment

This study presents **valuable** new insights into the patterns of organelle inheritance in the protozoan parasite *Toxoplasma gondii*. An innovative dual-labeling approach used in this study to track maternal-derived and de novo synthesized organelles provides a technical advance with potential to be more broadly applied. **Solid** evidence is provided that different organelles show distinct inheritance fates during cell replication; however, the data describing the residual body component in this process is **incomplete**.

<https://doi.org/10.7554/eLife.108685.1.sa3>

Abstract

Toxoplasma gondii replicates through endodyogeny, an unconventional form of internal budding in which two daughter cells are assembled within a single mother cell. During this process, daughter cells must acquire a full complement of organelles, which may be inherited from the mother, formed *de novo*, or assembled through a combination of both mechanisms. To date the fate of maternal components during replication remains poorly understood. We previously showed that F-actin-driven dynamics generate the intravacuolar network, which defines the residual body (RB) and facilitates recycling of microneme proteins. However, the inheritance and recycling of other organelles have not been systematically analysed.

To address this, we employed a dual HaloTag-based pulse-chase fluorescence labelling strategy to distinguish between *de novo*-synthesized and recycled proteins in replicating tachyzoites. This approach reveals three distinct organelle inheritance patterns: (1) direct transmission of intact maternal organelles (e.g., rhoptries, micronemes), (2) expansion and division of pre-existing maternal organelles with incorporation of newly synthesized components (e.g., Golgi apparatus, apicoplast), and (3) degradation of maternal structures without recycling (e.g., inner membrane complex). Furthermore, we identify Myosin F (MyoF) as the key motor protein that mediates the selective recycling of maternal organelles via the RB. These findings redefine the RB as an active trafficking hub and reveal a selective,

regulated system of organelle inheritance and recycling that is critical for intracellular organization and parasite development.

Introduction

Toxoplasma gondii tachyzoites replicate through a specialized form of internal budding known as endodyogeny, wherein two daughter cells (DCs) are assembled within the cytoplasm of the mother cell (Francia and Striepen, 2014 [↗](#); Hu et al., 2002 [↗](#)). This unique replication mechanism demands precise orchestration of organelle inheritance and spatial organization within the parasitophorous vacuole (PV). Apicomplexan parasites, including *T. gondii*, exhibit a highly polarized cellular architecture defined by specialized secretory organelles—micronemes, rhoptries, and dense granules—essential for host cell invasion and intracellular survival. These are accompanied by canonical organelles such as the Golgi apparatus, endoplasmic reticulum (ER), a single mitochondrion, and the apicoplast, a non-photosynthetic plastid. Notably, unlike most eukaryotes, *T. gondii* is enveloped by a pellicle consisting of the plasma membrane and an underlying inner membrane complex (IMC) (Ouologuem and Roos, 2014 [↗](#)), a structure composed of flattened vesicles tightly associated with the subpellicular cytoskeleton (Harding and Meissner, 2014 [↗](#)).

As DCs emerge during endodyogeny, they progressively encapsulate nearly all maternal cytoplasmic contents. While the sequence of organelle acquisition has been extensively characterized (Nishi et al., 2008 [↗](#)), the extent to which maternal organelles and proteins are recycled remains unclear. We previously showed that maternal micronemal proteins, such as MIC2, are almost entirely recycled into daughter cells via an F-actin-dependent mechanism and are trafficked through the intravacuolar network (IVN), which structurally organizes the residual body (RB) (Periz et al., 2019 [↗](#)). Historically viewed as a passive remnant of maternal cytoplasm, the RB has emerged as a key structure in coordinating replication, facilitating organelle recycling, and maintaining cytoplasmic continuity across individual parasites (Periz et al., 2017b [↗](#)). The RB forms at the posterior end of dividing parasites, physically linking them through the actin-rich IVN and enabling cytoplasmic exchange and synchronized development within the PV (Periz et al., 2017b [↗](#); Tosetti et al., 2019 [↗](#)).

Recent studies have shed some light on the molecular architecture and regulation of the IVN. We demonstrated that F-actin filaments organize the IVN, with disruption of actin (ACT1) resulting in parasite disorganization and impaired microneme recycling (Periz et al., 2019 [↗](#); Periz et al., 2017b [↗](#)). The RB and IVN also require the function of unconventional myosins MyoI and MyoJ, which are implicated in cell-cell communication (Frenal et al., 2017 [↗](#)). Additionally, the actin-nucleating factors Formin-2 and Formin-3 are required for IVN formation (Stortz et al., 2019 [↗](#); Tosetti et al., 2019 [↗](#)). However, the mechanisms driving the physical transport of maternal organelles remain poorly defined. F-actin could mediate this process either by serving as a track for myosin-dependent vesicular transport, as described for myosin VI in other eukaryotes (Frank et al., 2004 [↗](#)), or through the intrinsic mobility and dynamic association of actin bundles themselves, which could transiently interact to promote vesicle exchange (Das et al., 2021 [↗](#); Khaitlina, 2014 [↗](#); Moore et al., 2021 [↗](#)).

Although the biogenesis of daughter cell structures has been well described, the fate of maternal organelles during replication remains incompletely understood. Evidence suggests that maternal components such as MIC2 and the IMC protein GAP40 can be recycled via actin-dependent mechanisms (Ouologuem and Roos, 2014 [↗](#); Periz et al., 2019 [↗](#)). However, a systematic investigation of organelle recycling during endodyogeny has yet to be performed.

In this study, we employ a dual-labeling strategy using the HaloTag system (Urh and Rosenberg, 2012 [↗](#)) to investigate the fate of maternal organelles during *T. gondii* replication. Through fluorescence intensity-based analysis, we identify three distinct fates for maternal proteins: full

inheritance of intact organelles, incorporation of newly synthesized components into expanded maternal structures, and degradation without recycling. Furthermore, we demonstrate that the residual body functions as a key recycling hub, with this process dependent on the activity of the class XXII myosin F (MyoF). These findings provide new mechanistic insights into organelle inheritance and reveal how intracellular organization is maintained during parasite replication.

Results

HaloTag-Based pulse chase labelling allows discrimination of recycled and *de novo* material

To evaluate HaloTag as a tool for distinguishing maternal from newly synthesized proteins during endodyogeny, two marker proteins with known fates were used: MIC2, which undergoes extensive recycling (Periz et al., 2019 [↗](#)), and IMC1, primarily synthesized *de novo* (Ouologuem and Roos, 2014 [↗](#)). Using CRISPR/Cas9, endogenous loci were tagged to generate MIC2-Halo and IMC1-Halo strains (Singer et al., 2023 [↗](#)).

Both strains were sequentially labeled: maternal proteins were marked with Halo JF646 (M-MIC2, M-IMC1), followed 24 hours later by Halo JF549 to label newly synthesized proteins (n-MIC2, n-IMC1) (Figure 1A [↗](#)). Fluorescence intensity (FI) analysis across replication stages (Figure 1B–D [↗](#)) revealed a progressive decline of M-MIC2 and an increase in n-MIC2, confirming protein recycling. In contrast, M-IMC1 FI dropped from 100% to 14% between stages 1 and 2, with a corresponding rise in n-IMC1 to ~80%, indicating predominant *de novo* synthesis. These distinct FI dynamics demonstrate HaloTag's utility for resolving protein origin and support the *de novo* assembly of the IMC.

HaloTag profiling reveals three distinct patterns of protein inheritance

To further investigate protein inheritance during endodyogeny, endogenous HaloTag fusions were generated for a set of organelle-associated proteins: RON2 and ROP1 (rhoptries) (Besteiro et al., 2009 [↗](#); Striepen et al., 2001 [↗](#)), GRA1 (dense granules) (Carruthers and Sibley, 1997 [↗](#)), TIC20 (apicoplast) (Carruthers and Sibley, 1997 [↗](#)), SortLR (Golgi) (Sloves et al., 2012 [↗](#)), ANKER1 (ER) (Barylyuk et al., 2020 [↗](#)), GAPM1a (IMC) (Harding et al., 2019 [↗](#)), and MyoA (glideosome) (Meissner et al., 2002 [↗](#)) (Figure 2A [↗](#), Figure S1 [↗](#)).

Following the same sequential labeling strategy used for MIC2 and IMC1, fluorescence intensity (FI) of maternal proteins was quantified over successive replication cycles (Figure 2B, C [↗](#)). Analysis revealed three distinct inheritance profiles:

- **Group 1:** RON2 and ROP1 exhibited stable maternal FI, with minimal decline of FI within a single rhoptry, indicating recycling of whole organelles. In good agreement with this hypothesis, detailed analysis of M-RON2 inheritance demonstrated that the maternal rhoptries are separated during the distribution and that some daughter cells did not obtain maternal organelles after successive round of replication (Figure 2 [↗](#), Figure S2 [↗](#)).
- **Group 2:** Proteins from diverse organelles (SortLR, ANKER1, TIC20, MyoA) showed a stepwise ~50% reduction in maternal FI per division, suggesting that recycled, maternal material and *de novo* material end up in the same organelle
- **Group 3:** IMC proteins (IMC1 and GAPM1a) displayed a sharp FI loss between the first and second replication cycles, consistent with predominant *de novo* synthesis of the IMC and degradation of maternal material.

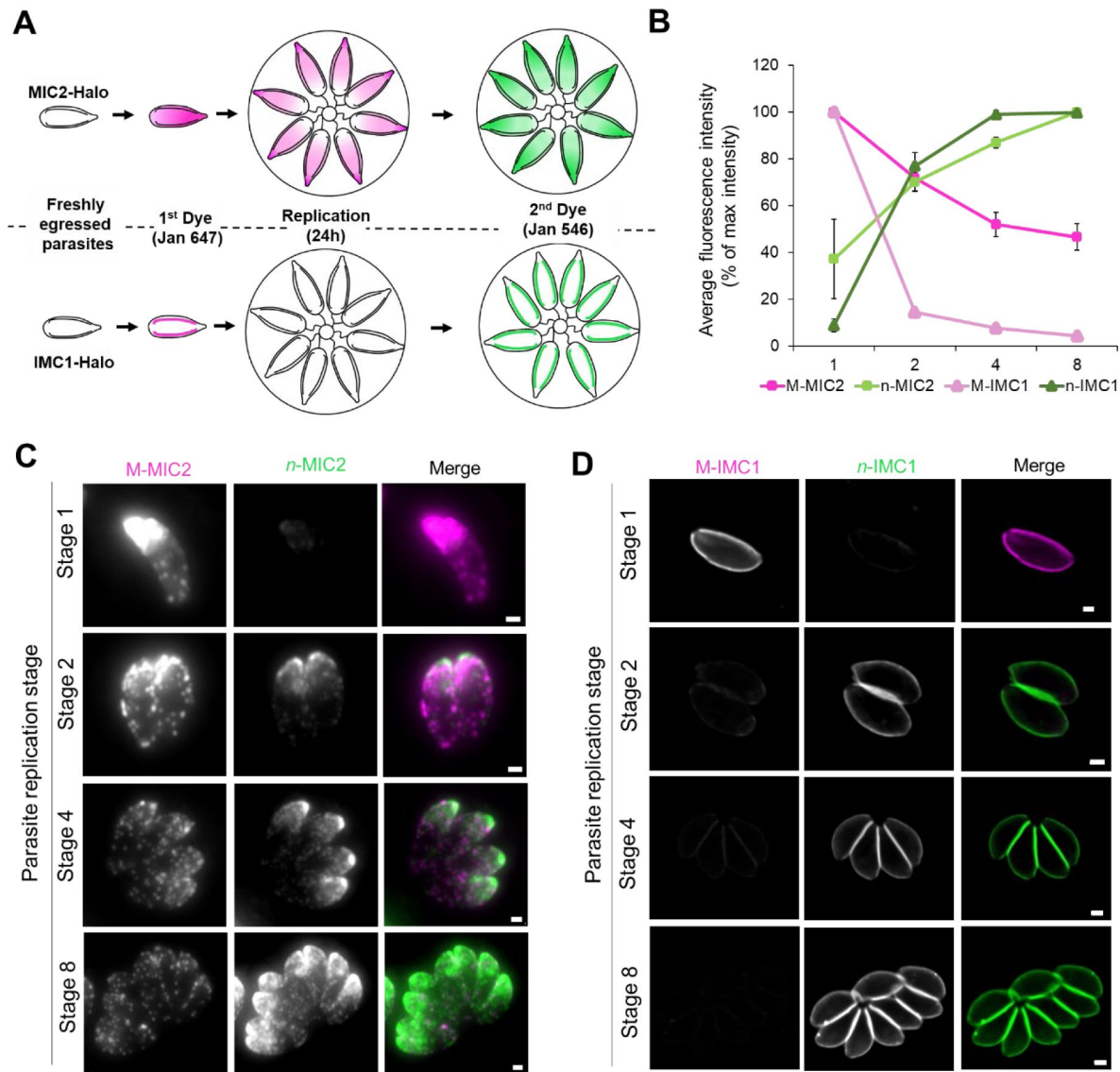


Figure 1

Discrimination between maternal and *de novo* material.

A) Dual staining scheme. Maternal material is labelled initially with Janila Fluor®-646, and the excess washed out. After replication of parasites *de novo* synthesised material is labelled with Janila Fluor®-549, allowing efficient discrimination of the two populations. **B**) Fluorescence intensity quantification of maternal (M) and *de novo* synthesised (n) MIC2 and IMC1 during replication from stage 1 to 8. Magenta: M-MIC2, Rosa: M-IMC1, Green: n-MIC2, Dark Green: n-IMC1. **C**) Representative picture of MIC2-Halo parasites during replication (stage 1 to 8) stained for both maternal (M) and *de novo* (n) MIC2. Magenta: M-MIC2, Green: n-MIC2. Maternal MIC2 is efficiently recycled into the daughters, while *de novo* MIC2 is formed after each replication cycle **D**) Representative picture of IMC1-Halo parasites during replication (stage 1 to 8) stained for both maternal (M) and *de novo* (n) IMC1. Magenta: M-IMC1, Green: n-IMC1. In contrast to micronemes, maternal IMC is degraded, and only residual amount is detected after the first replication cycle. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.

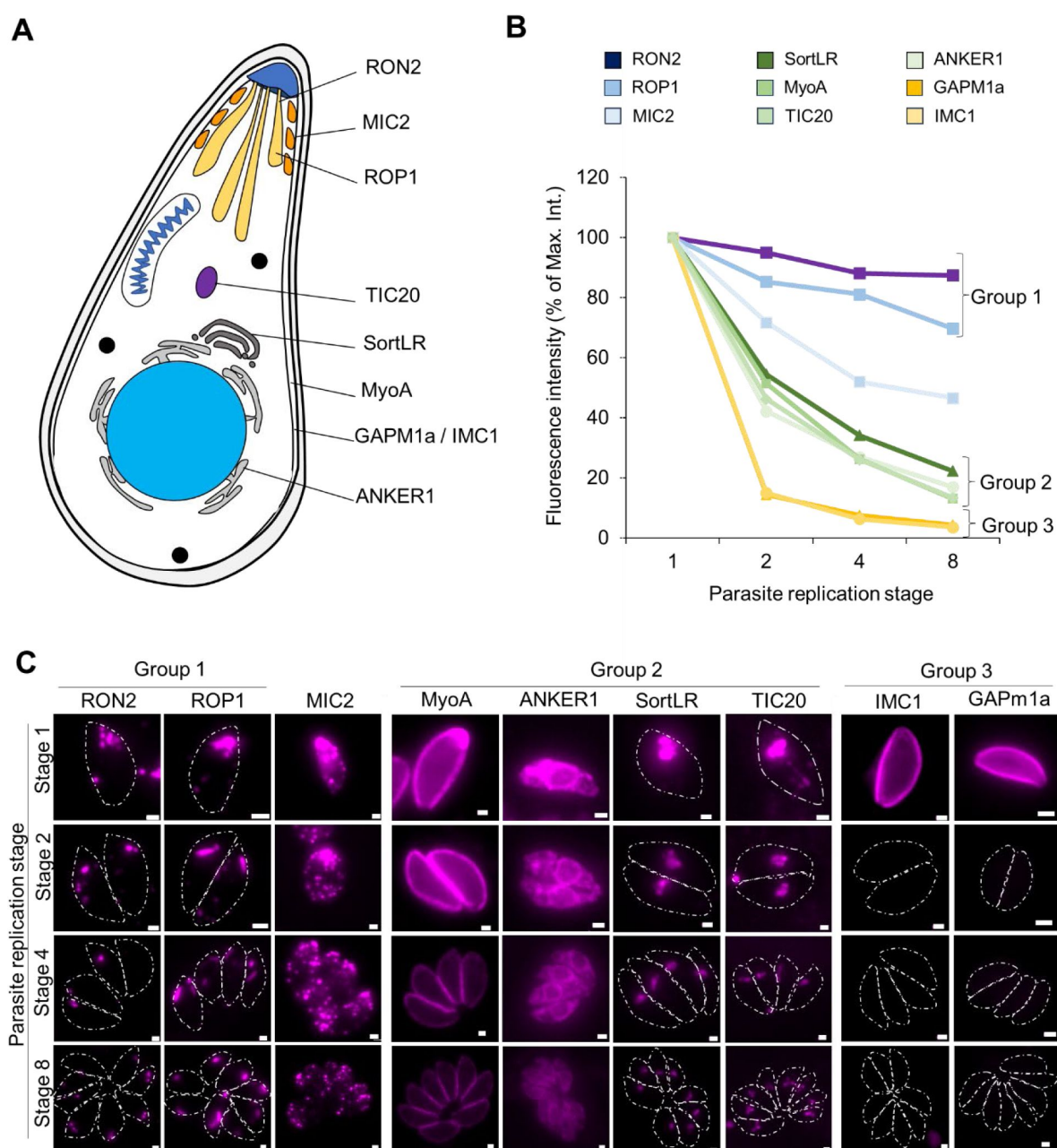


Figure 2

Three recycling fates for the maternal organelles.

A) Overview of *T. gondii* organelles and molecular markers used for visualisation with endogenous Halo-Tags. **B)** Fluorescence intensity quantification of maternal (M) molecular marker listed in A. The fluorescence intensity variation during replication, allows to group them into three groups: 1) Efficient, almost quantitative recycling. 2) even distribution of maternal material and 3) almost exclusive *de novo* synthesis. Dark blue: RON2, Blue: ROP1, Pale blue: MIC2, Dark green: SortLR, Green: MyoA, Pale green: Tic20, Greenish white: ANKER1, Gold: GAPM1a, Yellow: IMC1. **C)** Representative picture of parasites expressing indicated Halo tagged proteins (stage 1 to 8) stained for maternal (M) proteins. Three biological replicates were used for all analyses; the graph indicate the mean value of the triplicate. All scale bars = 1 μ m.

Interestingly, MIC2 exhibited intermediate behavior, prompting further analysis to refine its classification. These patterns highlight distinct organelle-specific protein inheritance mechanisms in *T. gondii*.

Micronemes and rhoptries are recycled as intact organelles during *Toxoplasma* endodyogeny

Micronemes are known to comprise distinct subpopulations (Kremer et al., 2013 [↗](#)), and prior work has shown that recycled and *de novo* MIC2 localize to separate microneme subsets (Periz et al., 2019 [↗](#)). Based on these observations, we hypothesized that both micronemes and rhoptries are recycled as intact organelles, with new organelles formed independently via *de novo* synthesis.

To test this, we conducted detailed analyses of MIC2 and rhoptry inheritance. As expected, recycled (M-MIC2) and newly synthesized MIC2 (n-MIC2) were largely segregated into distinct micronemes with minimal colocalization (Figure S3 [↗](#)) (Periz et al., 2019 [↗](#)). Focusing on sparsely distributed micronemes outside the apical region (Figure 3A [↗](#), boxed area), we measured fluorescence intensity (FI) of individual M-MIC2-positive micronemes at replication stages 2, 4, and 8 (Figure 3A [↗](#), zoomed region; 3C). The relatively stable FI of these isolated structures supports whole organelle recycling, consistent with the FI profiles of rhoptries (Figure 2B, C [↗](#)).

Rhoptry analysis further corroborated this model. As with MIC2, M-RON2 and n-RON2 localized to distinct rhoptry populations (Figure S2 [↗](#)), with divergence becoming more apparent at stage 8 (Figure 3D [↗](#), Figure S2D [↗](#)). M-RON2 FI remained relatively constant through replication (Figure 3F [↗](#)), while n-RON2 gradually increased from stage 2 onward (Figure S2C [↗](#)). Due to their low copy number (8–12 per tachyzoite), maternal rhoptries were separated and often absent in some daughter cells after successive divisions, consistent with stochastic whole-organelle inheritance illustrated by decrease of the M-RON2 fluorescence signal average area and the constancy of the total surface area (Figure S2A, B [↗](#)). As consequence a progressive decline in M-RON2-positive parasites per vacuole can be observed (Figure 3E, D [↗](#); Figure S2D [↗](#)). In contrast, micronemes showed a more controlled segregation: even in large parasitophorous vacuoles, all daughter parasites retained a mix of recycled and newly synthesized micronemes (Figure S4 [↗](#)), indicating regulated partitioning.

Together, these results demonstrate that both micronemes and rhoptries are recycled as intact organelles. *De novo* organogenesis acts in parallel to maintain a full complement in each daughter cell, ensuring faithful inheritance during endodyogeny.

Golgi inheritance involves coordinated expansion and partitioning of maternal and *de novo* proteins

Next, we analyzed SortLR, a Golgi marker, using dual HaloTag labeling. Maternal SortLR (M-SortLR) fluorescence declined by ~50% per replication round, while newly synthesized SortLR (n-SortLR) increased proportionally, without abrupt loss (Figure 4A–B [↗](#)). This gradual dilution contrasts sharply with the rapid decay seen in Group 3 proteins.

M-SortLR and n-SortLR exhibited strong colocalization across replication stages, and all daughter cells retained M-SortLR (Figure 4C [↗](#)). The total Golgi area expanded from ~2 μm^2 to ~10 μm^2 by stage 8 (Figure 4D [↗](#)), consistent with coordinated Golgi elongation and medial fission, as described previously (Pelletier et al., 2002 [↗](#)). These results demonstrate that the Golgi undergoes autonomous duplication, incorporating both maternal and *de novo* proteins during its growth and partitioning.

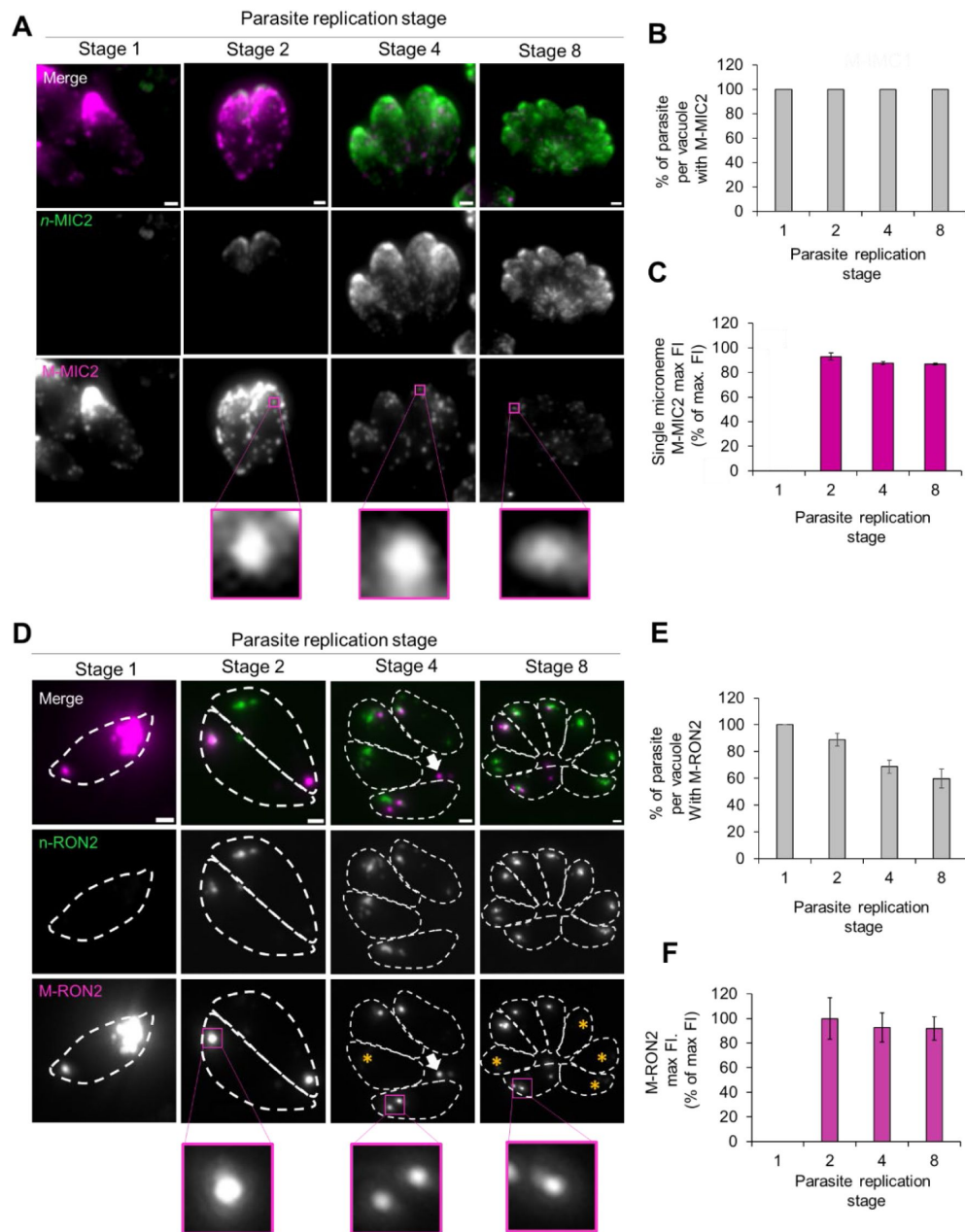


Figure 3

Micronemes and rhoptries are recycled as whole organelles

A) Representative picture of MIC2-Halo during replication (stage 1 to 8) stained for both the maternal (M) and *de novo* (n) MIC2. Magenta: M-MIC2, Green: n-MIC2. Zoom windows allow to visualise isolated apical M-MIC2 signal for which the fluorescence intensity is quantified in C. **B**) Quantification of the inheritance of M-MIC2 by the daughter cells during replication (stage 1 to 8). All daughter cells inherited M-MIC2. **C**) Fluorescence intensity quantification of isolated M-MIC2 signal as illustrated in A. **D**) Representative picture of RON2-Halo during replication (stage 1 to 8) stained for both the maternal (M) and *de novo* (n) MIC2. Magenta: M-RON2, Green: n-RON2. Zoom windows allow to visualise isolated apical M-RON2 signal for which the fluorescence intensity is quantified in F. **E**) Quantification of the inheritance of M-RON2 by the daughter cells during replication (stage 1 to 8). Not all daughter cells inherit M-RON2 (asterisks), best seen in later replication stages. **F**) Fluorescence intensity quantification of isolated M-RON2 signal as illustrated in D. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.

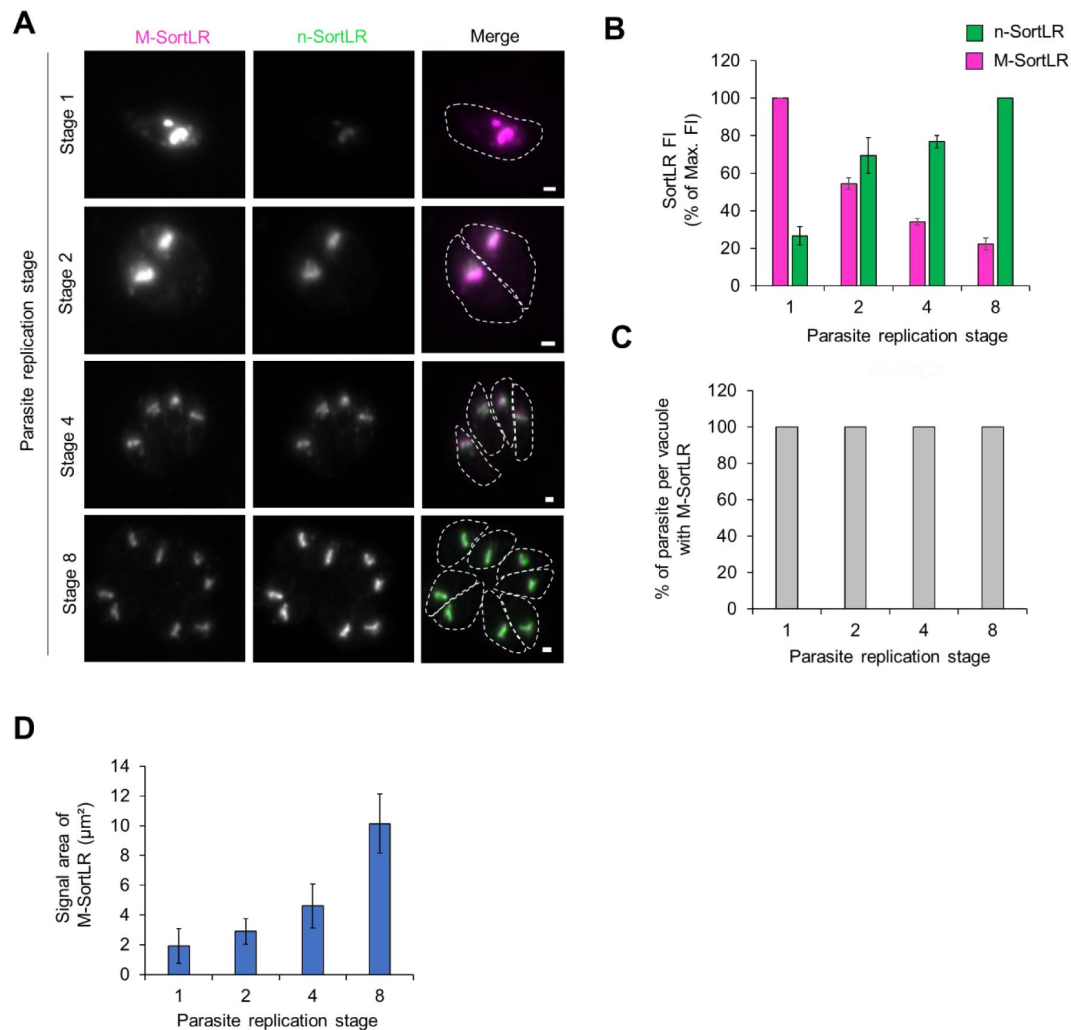


Figure 4

Expansion of the mother organelle for equivalent sharing to the daughter.

A) Representative picture of SortLR-Halo during replication (stage 1 to 8) stained for both the maternal (M) and *de novo* (n) SortLR. Magenta: M-SortLR, Green: n-SortLR. **B)** Fluorescence intensity quantification of M-SortLR and n-SortLR illustrated in A. Magenta: M-SortLR, Green: n-SortLR. **C)** Quantification of the inheritance of M-SortLR by the daughter cells during replication (stage 1 to 8). All daughter cells inherited M-SortLR. **D)** Quantification of the signal area of M-SortLR during replication. The total surface of M-SortLR increase at each replication by about a factor 2. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μm .

Thus, Group 2 organelles follow a distinct inheritance mode, involving regulated expansion, integration of new material, and equal distribution to progeny—distinct from the whole-organelle recycling of rhoptries/micronemes and the complete turnover seen in IMC proteins.

Maternal IMC proteins are degraded in the RB during replication

Group 3 proteins, typified by IMC components like GAPM1a and IMC1, exhibit a sharply distinct inheritance profile, characterized by rapid loss of maternal fluorescence during replication. Using HaloTag-based dual labeling, we observed a ~90% reduction in maternal GAPM1a signal by stage 2, with minimal retention through later stages (**Figure 5A–C**), indicative of active degradation rather than recycling. M-GAPM1a and n-GAPM1a did not colocalize, and residual maternal signal localized transiently at the posterior pole before disappearing entirely, suggesting disposal via the residual body (RB) (**Figure 5D**).

Live-cell imaging of IMC1 further supported this model: maternal IMC1 (M-IMC1) condensed at the posterior pole during daughter budding, followed by a marked fluorescence drop, without redistribution to other regions (**Figure 5E, F** and **Figure S5**, **Video S1**). This signal loss was replication-specific, as non-dividing parasites retained stable M-IMC1 levels (**Figure S5D**). Quantification across regions revealed a transient increase at the posterior, followed by complete disappearance, consistent with degradation (**Figure 5F**, **Figure S5C**).

Interestingly, daughter IMCs (DCs) exhibited brighter fluorescence than the maternal IMC (**Figure S5E, F**), especially during early stages of elongation (**Figure S5 G, H**). FI peaked when DCs reached ~3 μm in length, suggesting the IMC is fully assembled before emergence and subsequently unfolds without requiring additional material. This preformation explains the absence of maternal IMC recycling. These findings align with earlier observations that IMC components are synthesized de novo during elongation (Oueloguem and Roos, 2014).

Distinct recycling pathways govern whole-organelle inheritance of rhoptries and micronemes

Our data suggest that the residual body (RB) plays a selective role in organelle recycling: while microneme and rhoptry proteins are routed through the RB, IMC proteins are primarily degraded within this compartment. In contrast, recycling of maternal Golgi, ER, and apicoplast proteins appears to occur directly, bypassing the RB.

To investigate this further, we performed live-cell imaging of rhoptry and microneme inheritance (**Figure 6A, B**, **Video S2, S3**). MIC2-Halo parasites co-expressing IMC1-YFP, showed that maternal MIC2 (M-MIC2) transits through the RB (**Figure 6A**, white arrows). Notably, after endodyogeny was completed, M-MIC2 was redistributed from the RB to the apical tip of the daughter cells (**Figure 6A**, 11:30, 16:00 h), confirming that the RB serves as a temporary reservoir during microneme recycling (Periz et al., 2019).

For rhoptries, we examined the inheritance of maternal RON2 (M-RON2) in RON2-Halo parasites co-expressing IMC1-YFP. In approximately 90% of parasites undergoing replication, M-RON2 was integrated into daughter rhoptries prior to mother cell collapse and formation of the RB (**Figure 6B**, 4:15–4:30 h and 10:30–10:45 h). Like M-MIC2, M-RON2 was occasionally detected in the RB, though less prominently, suggesting more rapid or tightly regulated recycling.

To test whether rhoptry recycling also depends on F-actin, as previously reported for micronemes (Periz et al., 2019), we expressed the F-actin marker Cb-Emerald (Periz et al., 2017b) in the RON2-Halo line (**Figure 6C, D**). While m-MIC2 displayed clear F-actin-associated movement along filaments in the residual body (**Figure 6C**), colocalization of M-RON2 with Cb-Emerald in the residual body was infrequent. On the other hand, both secretory organelles were found to be associated with cytoplasmic F-actin (**Figure 6D**).

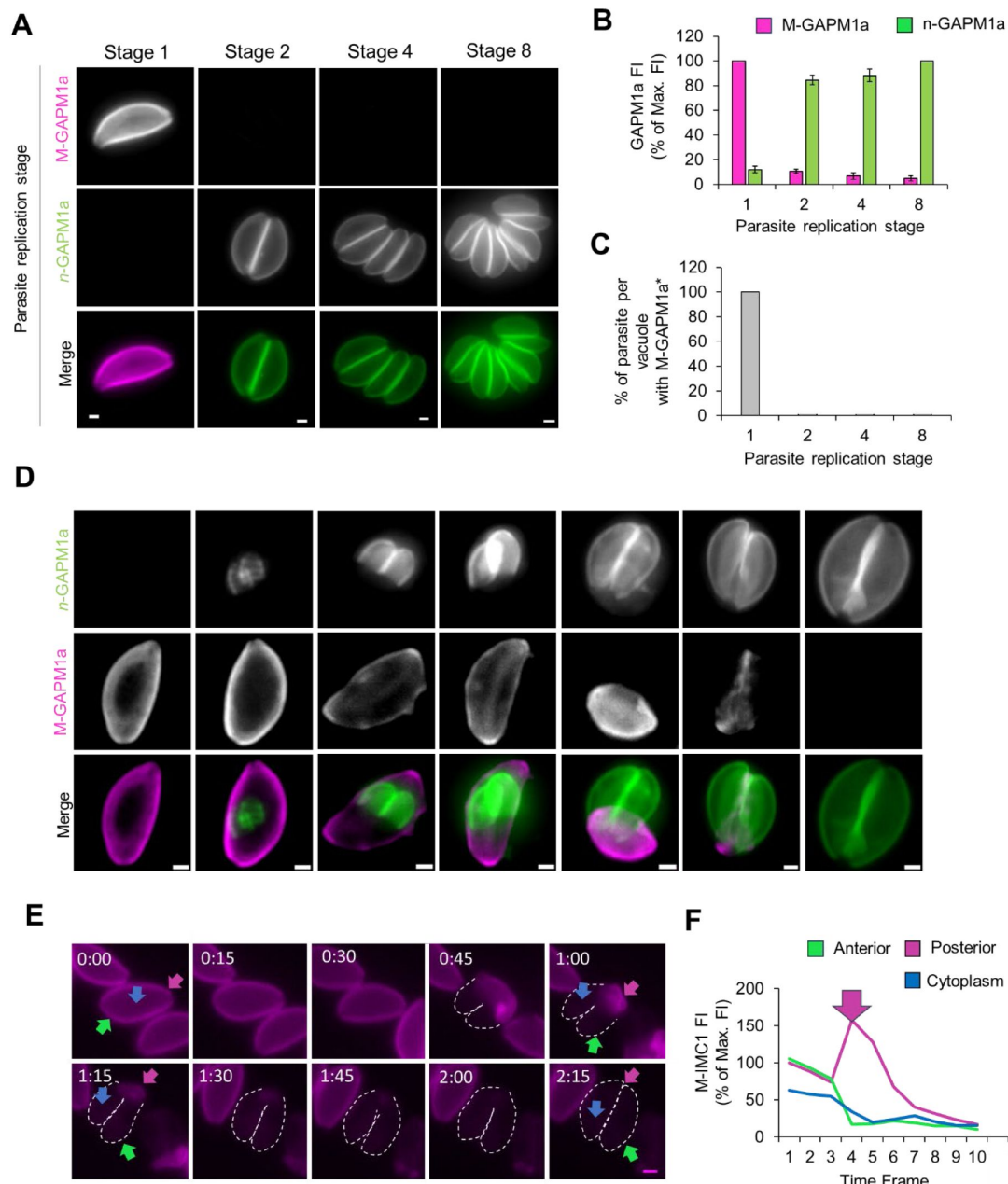


Figure 5

Degradation of the inner membrane complex.

A) Representative picture of GAPM1a-Halo during replication (stage 1 to 8) stained for both the maternal (M) and *de novo* (n) GAPM1a. **B**) Fluorescence intensity quantification of M-GAPM1a and n-GAPM1a illustrated in A. **C**) Quantification of the inheritance of M-GAPM1a by the daughter cells during replication (stage 1 to 8). After replication, M-GAPM1a is not visible in daughter cells. **D**) GAPM1a is degraded in the RB. Representative images of GAPM1a-Halo parasites at different stages of daughter cell development. M-GAPM1a collapses toward the forming residual body, where the signal disappears after completion of replication, indicating its degradation. **E**) Time series of IMC1-Halo parasites during the first replication. Parasites were stained for the maternal (M) IMC1 prior invasion. Three regions of the parasites were analysed: the apical (green arrow), the cytoplasmic (blue arrow) and the basal (magenta arrow). **F**) Fluorescence intensity analysis of the three regions defined in E. The curve and analysed region arrow share the same colour code. Green: Apical, Blue: Cytoplasm, Magenta: Basal. After the accumulation of the mother IMC at the basal pole, the FI of M-IMC1 decrease without redistributing to any other region. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.

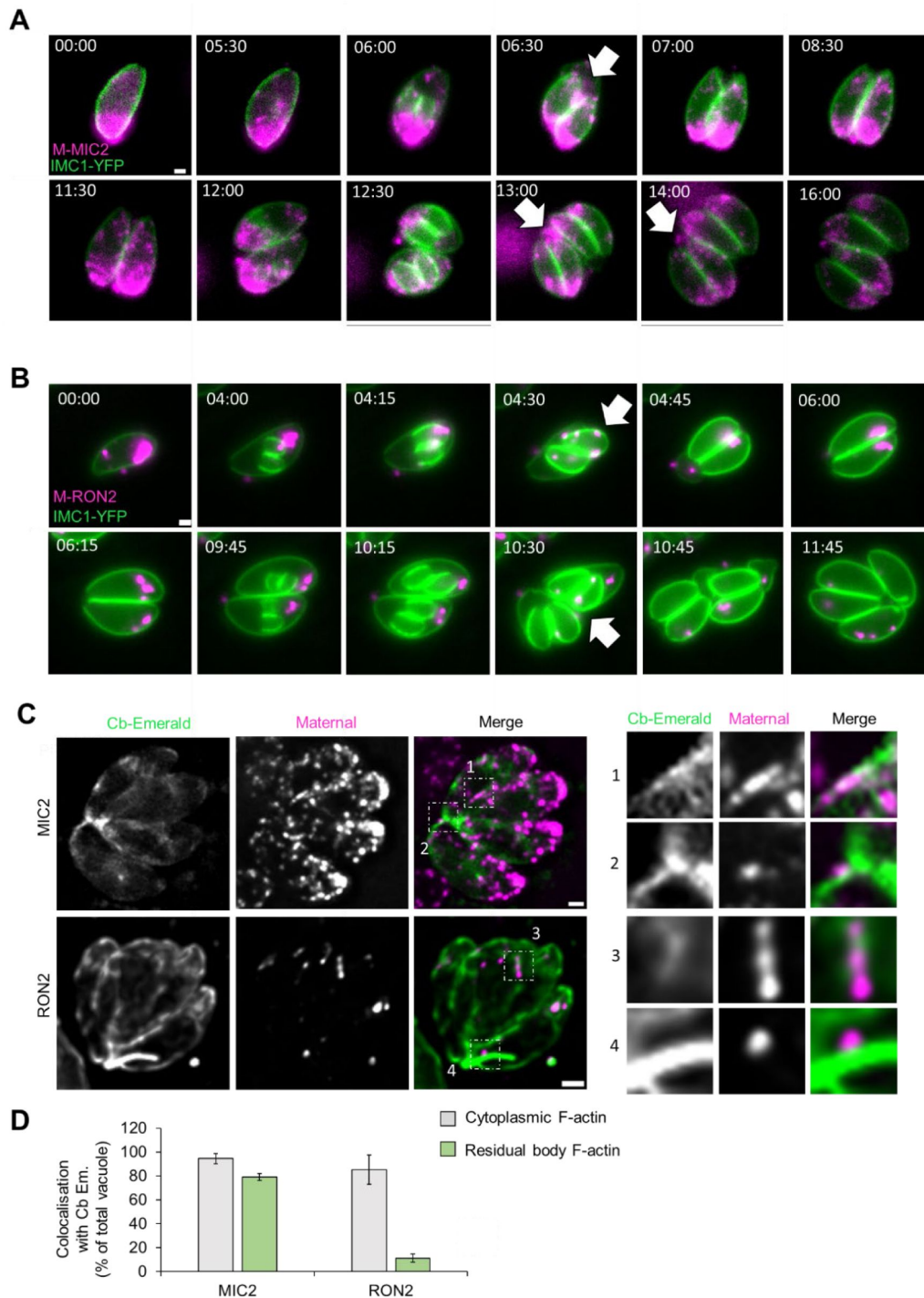


Figure 6

Micronemes and rhoptries are recycled similarly, but differ in timing and location.

A) Time-lapse of MIC2-Halo over two replication cycles. Maternal MIC2 (M-MIC2) is stained, and daughter cell formation is visualized via IMC1-YFP. M-MIC2 is transported via the residual body (RB) (white arrows; see Video S2 [\[link\]](#)). B) Time-lapse of RON2-Halo parasites with IMC-YFP. Maternal RON2 (M-RON2) is transported before mother cell collapse and RB formation (white arrows). C) Representative images showing maternal organelles associating with F-actin and the RB. Insets highlight colocalization of maternal proteins and F-actin. D) Quantification of M-MIC2 and M-RON2 colocalization with F-actin. Data from three biological replicates; bars represent means $\pm$ SD. Scale bars = 1 μ m. *Myosin-F drives F-actin mediated recycling of maternal organelles via the residual body.*

In summary, both rhoptries and micronemes are inherited as intact organelles during *T. gondii* replication. However, the underlying timing differs: microneme recycling is delayed until after daughter emergence, whereas rhoptry inheritance is more direct and occurs prior to mother cell collapse. These findings highlight the organelle-specific complexity of recycling pathways and support whole-organelle recycling of micronemes and rhoptries.

Given the F-actin dependency of this recycling pathway (Periz et al., 2019 [↗](#)), we hypothesized that a conserved apicomplexan myosin mediates this process. MyoF, a class XXII myosin is conserved across Apicomplexa (Jacot et al., 2013 [↗](#)) and has been implicated in diverse functions—from apicoplast inheritance to F-actin-dependent vesicular transport (Carmeille et al., 2021 [↗](#); Heaslip et al., 2016 [↗](#); Kellermeier and Heaslip, 2024 [↗](#)).

We endogenously Halo-tagged MIC2, RON2, and SortLR in the auxin-inducible degron line MyoF-mAID (Carmeille et al., 2021 [↗](#)) and performed dual HaloTag labelling to differentiate maternal (M) from de novo (n) protein pools.

Live imaging revealed that MyoF depletion resulted in the accumulation of maternal but not *de novo* MIC2 and RON2 in the RB during early replication stages (**Figure 7A, B** [↗](#)). At later stages, recycled *de novo* organelles from earlier cycles (now maternal) also began to accumulate. Immunofluorescence with α -AMA1, α -MIC4, and α -MIC8 confirmed that this phenotype broadly affects the entire repertoire of micronemal proteins (**Figure S6** [↗](#), **S7** [↗](#)). Quantification showed that approximately 90% of vacuoles exhibited RB accumulation of maternal material upon auxin treatment (**Figure 7D** [↗](#)), indicating that MyoF specifically mediates the recycling of maternal micronemes and rhoptries. Interestingly, although our time lapse analysis indicated that rhoptries only rarely traffic via the RB, upon depletion of MyoF, they are accumulating in the RB, supporting the hypothesis that micronemes and rhoptries are transported in the same manner during recycling.

In contrast, Golgi inheritance (marked by SortLR) remained unaffected by MyoF depletion (**Figure 7C** [↗](#)). Neither accumulation in the RB nor separation of maternal and *de novo* SortLR signals was observed, consistent with previous reports that Golgi duplicates and partitions independently (Pelletier et al., 2002 [↗](#)).

To confirm that this phenotype results directly from MyoF depletion and not broader disruption of endomembrane architecture, we performed a rescue experiment. MIC2-Halo MyoF-mAID parasites were cultured in auxin for 24 hours to induce M-MIC2 retention in the RB, followed by auxin wash-off and continued replication (**Figure 8A** [↗](#)). Remarkably, M-MIC2 was redistributed to daughter cells in the following cycles, occasionally in an asymmetric fashion (**Figure 8B** [↗](#)), possibly reflecting unequal MyoF re-expression across daughters.

Live-cell imaging confirmed these dynamics (**Video S4, S5** [↗](#)). In untreated parasites, M-MIC2 entered the RB ~7 h post-replication and was later redistributed to daughter cells (**Figure 8D** [↗](#), **Video S4** [↗](#)). Under auxin treatment, M-MIC2 accumulated post-budding and failed to redistribute (**Figure 8E** [↗](#); **Video S5** [↗](#)). Wash-off rescued redistribution during the second cycle, though visualization required intensity adjustments due to signal differences between accumulated and redistributed pools (**Figure 8F** [↗](#); **Video S6** [↗](#)).

Together, these findings establish that MyoF is essential for recycling maternal micronemes and rhoptries via the RB, redefining the RB as a dynamic trafficking hub, not merely a degradative compartment. MyoF acts as a mechanistic bridge between F-actin dynamics and organelle inheritance, orchestrating precise redistribution of maternal cargo.

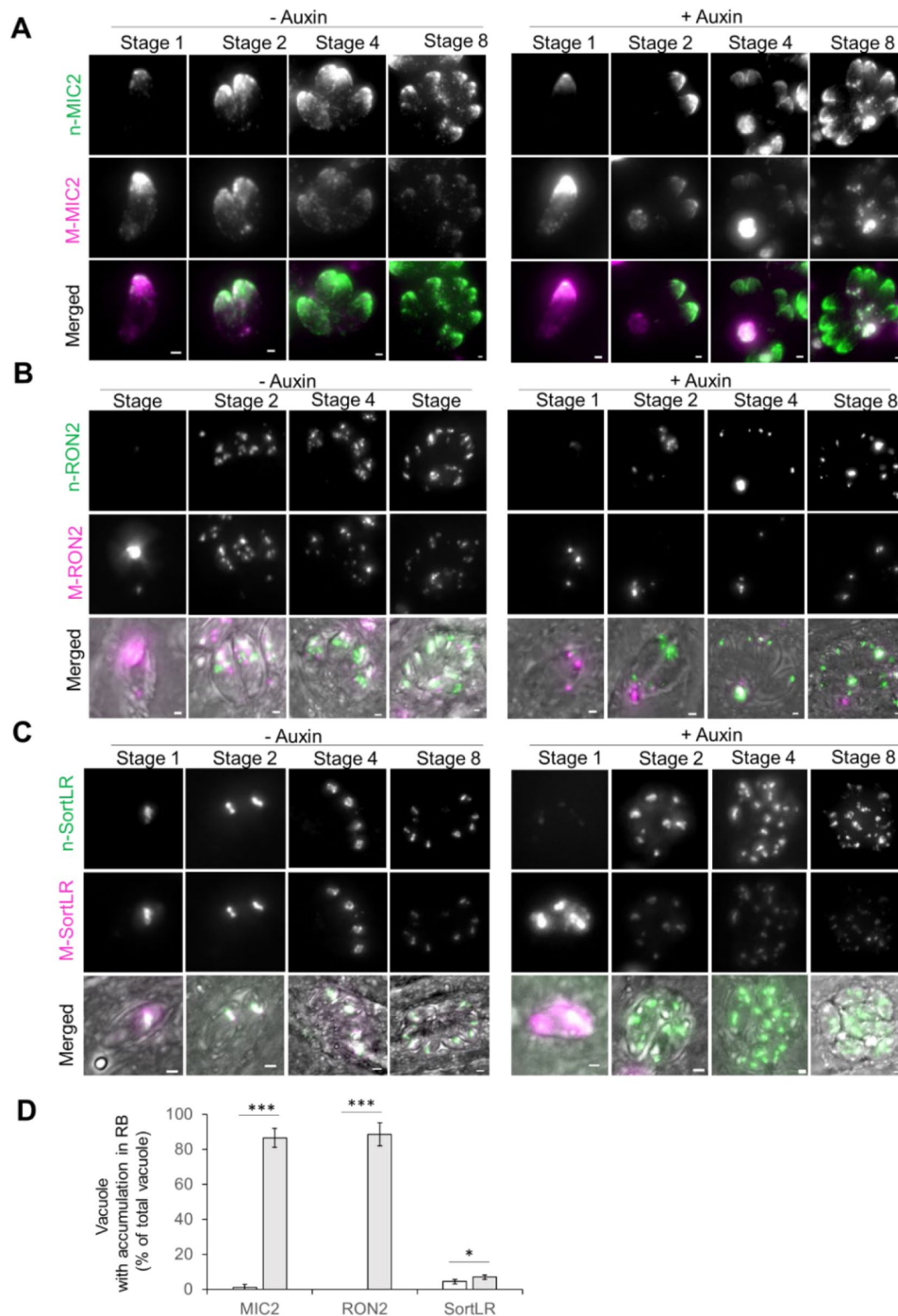


Figure 7

MyoF is essential for recycling apical secretory organelles.

A) Effect of MyoF knockdown (KD) on maternal microneme inheritance. MyoF-mAID MIC2-Halo parasites were labeled and grown $\pm$ auxin for 24h. Magenta: maternal MIC2 (M-MIC2), Green: newly synthesized MIC2 (n-MIC2). Without MyoF, M-MIC2 accumulates in the residual body (RB) instead of being passed to daughter cells (Stage 1–2), with increasing accumulation over replication cycles (Stage 2–8). B) Effect of MyoF KD on maternal Golgi inheritance. MyoF-mAID RON2-Halo parasites show M-RON2 retention in the RB, mirroring the pattern seen with MIC2. C) Effect of MyoF KD on maternal Golgi inheritance. MyoF-mAID SortLR-Halo parasites show no significant RB accumulation of M-SortLR, unlike MIC2 and RON2. D) Quantification of vacuoles showing RB accumulation. White: control; Gray: MyoF-KD. Three biological replicates were used; bars show means $\pm$ SD. All p-values $\leq$ 0.001 (***), using one-tailed unpaired Student's t-test. Scale bars = 1 μ m

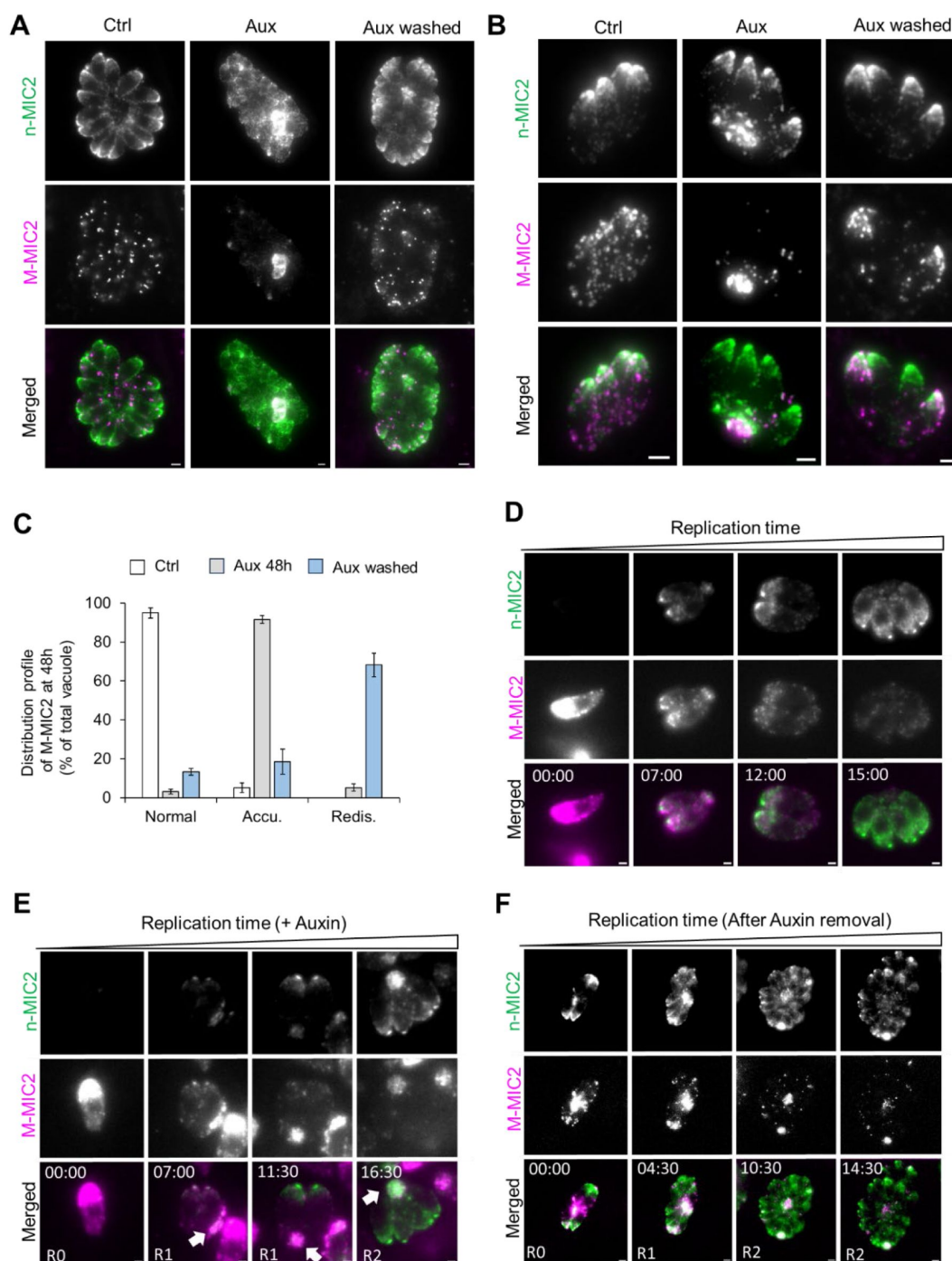


Figure 8

The residual body (RB) functions as a recycling center, not a dead end.

To assess the fate of material accumulated in the RB, auxin chase experiments were performed: 24h with auxin followed by 24h without. A) Representative images of MyoF-mAID MIC2-Halo parasites after 48h in control (no auxin), continuous auxin (Aux), or auxin washout (Aux washed). Magenta: M-MIC2, Green: n-MIC2. Continuous auxin led to M-MIC2 accumulation in the RB, while auxin washout enabled redistribution. B) Close-up of heterogeneous M-MIC2 redistribution post-auxin washout. C) Quantification of vacuoles showing normal, accumulated, or redistributed M-MIC2. White: control; Gray: MyoF-KD; Blue: MyoF-KD auxin chase. D) Time-lapse without auxin: M-MIC2 briefly passes through the RB (white arrows). E) Time-lapse with auxin: sustained M-MIC2 accumulation in the RB throughout replication. F) Time-lapse after auxin washout: initial M-MIC2 accumulation in the RB resolves by the second replication cycle, with redistribution observed. Three biological replicates were used; bars show means $\pm$ SD. Scale bars = 1 μ m.

Discussion

Use of dual labelling to differentiate maternal and de novo material

Understanding the complexity of organelle inheritance in *Toxoplasma gondii* requires tools capable of resolving both the spatial and temporal dynamics of protein trafficking. Conventional single-labelling approaches have been limited to static snapshots, obscuring key processes such as organelle biogenesis, recycling, and selective degradation. To overcome these challenges, we employed a dual HaloTag pulse-chase labelling strategy to distinguish maternally inherited from de novo-synthesized proteins. This approach, previously underutilized in apicomplexan parasites (Koreny et al., 2023 [↗](#); Periz et al., 2019 [↗](#)), allowed us to resolve protein fates across replication cycles and map distinct inheritance routes for multiple organelles.

The residual body as a regulatory hub

Our findings reaffirm the emerging view of the RB as an active, multifunctional structure rather than a passive cytoplasmic remnant (Frenal et al., 2017 [↗](#); Periz et al., 2017a [↗](#)). First described over 50 years ago (Sheffield and Melton, 1968 [↗](#)), the RB was long believed to serve as a repository for discarded material, often called a “waste bin” (Nishi et al., 2008 [↗](#)). However, we show that the RB temporarily stores maternal secretory organelles, such as micronemes and rhoptries, which are later redistributed to daughter cells in a MyoF-dependent manner.

Live-cell imaging revealed that organelles in the RB are not static but can re-enter functional pathways (Figures 8–10 [↗](#)), establishing the RB as a trafficking checkpoint (Periz et al., 2019 [↗](#)). This active role is further supported by the dependence of RB-mediated recycling on F-actin and the class XXII myosin MyoF, which we show to be essential for retrieval of maternal MIC2 and RON2 but dispensable for Golgi inheritance although we noticed a fragmentation of the Golgi, which has been described to depend on MyoF (Carmeille et al., 2021 [↗](#)).

The RB is also critical for cytoplasmic continuity across parasites, synchronizing replication and facilitating inter-parasite protein exchange (Frenal et al., 2017 [↗](#); Muniz-Hernandez et al., 2011 [↗](#); Periz et al., 2017a [↗](#)). Its formation is dependent on actin-nucleating factors and myosins, indicating a complex and regulated origin rather than a passive leftover of cytokinesis (Stortz et al., 2019 [↗](#); Tosetti et al., 2019 [↗](#)). Given these parallels, the RB may be functionally analogous to the mammalian midbody remnant, which regulates post-mitotic signaling and intracellular trafficking (Kuriyama et al., 2025 [↗](#); Peterman and Prekeris, 2019 [↗](#)).

Fate determination: inheritance or degradation?

An important question raised by our study is how the parasite determines whether maternal proteins are recycled or degraded. Our results indicate that this decision is selective rather than stochastic (Figure 9 [↗](#)). For instance, maternal micronemes persist for at least six replication cycles in the absence of MyoF, without signs of degradation. In contrast, IMC proteins like GAPM1a and IMC1 are lost within a single cycle and show no evidence of recycling. Degradation appears to occur within the RB itself, raising questions about the underlying mechanism. Although *T. gondii* encodes homologs of lysosomal and autophagic degradation machinery (Besteiro et al., 2011 [↗](#); Smith et al., 2021 [↗](#); Thaprawat et al., 2025 [↗](#)), none have been localized to the RB or directly linked to IMC degradation. However, ubiquitin ligases and proteasomal components have been identified in the RB proteome (O’Shaughnessy et al., 2023 [↗](#); Que et al., 2002 [↗](#)). IMC proteins are also part of the ubiquitinated proteome (Silmon de Monerri et al., 2015 [↗](#)), and recent work showed that deletion of the kinase ERK7 prevents sequestration of the E3 ligase CSAR1, resulting in

aberrant degradation of conoids (O'Shaughnessy et al., 2023 [DOI](#)). These findings support a model in which protein fate within the RB is determined through selective ubiquitination and regulated proteolysis.

A call to reassess recycling factors in *Toxoplasma gondii*

The recognition of the residual body (RB) as a key site for organelle recycling has fundamentally revised our understanding of intracellular trafficking in *T. gondii* (Periz et al., 2019 [DOI](#); Tosetti et al., 2019 [DOI](#)). Earlier models emphasized *de novo* synthesis of organelles and proposed a repurposing of classical endocytic machinery for secretory functions (Tomavo, 2014 [DOI](#)). However, our HaloTag-based pulse-chase studies show that secretory organelles such as micronemes and rhoptries are extensively recycled via the RB, in a process dependent on F-actin and the unconventional myosin MyoF.

This shift in our understanding calls for a systematic re-evaluation of trafficking factors traditionally implicated in protein targeting, vesicle transport, and organelle biogenesis. Many of these particularly Rab-GTPases, SNAREs, and dynamins were previously studied without consideration of recycling pathways. Notably, accumulation of microneme material in the RB has been observed upon perturbation of several trafficking regulators, including Rab5A (Kremer et al., 2013 [DOI](#)), ArlX3 (Klinger et al., 2024 [DOI](#)), SORTLR (Sloves et al., 2012 [DOI](#)), VPS8 (Morlon-Guyot et al., 2018 [DOI](#)), and VPS11 (Morlon-Guyot et al., 2015 [DOI](#)). Yet, whether these phenotypes reflect a block in secretion, synthesis, or recycling remains unclear.

To resolve this, a targeted reanalysis of trafficking factors using our dual-labelling HaloTag assay combined with the splitCas9 system (Li et al., 2022 [DOI](#)) would allow high-throughput functional screening to specifically distinguish defects in recycling from those in *de novo* biogenesis, offering a path to redefine the roles of classical and lineage-specific trafficking regulators in *T. gondii*.

Material and Methods

Parasite culture and genetic manipulation

Growth and generation of transgenic *T. gondii*

T. gondii tachyzoites from the RH strain and derived lines, including RH Δ ku80/TATi and RH Δ ku80/Tir1, were maintained at 37 °C with 5% CO₂ in human foreskin fibroblasts (HFFs; ATCC, SCRC 1041) cultured in Dulbecco's Modified Eagle Medium (DMEM; Sigma, D6546) supplemented with 10% fetal bovine serum (FBS; BioSell, FBS.US.0500), 4 mM L-glutamate (Sigma, G7513), and 20 µg/mL gentamycin (Sigma, G1397) as previously described (Gras et al., 2019 [DOI](#)).

Generation of transgenic parasites

New strains were generated using CRISPR/Cas9 as previously described (Li et al., 2022 [DOI](#)). Guide RNAs (gRNAs) targeting the regions of interest were designed using EuPaGDT (Alvarez-Jarreta et al., 2024 [DOI](#)). All gRNA and primer sequences are listed in Supplementary Table 1. Briefly, gRNA oligos were annealed, ligated into the Cas9-YFP vector, and verified by sequencing (Eurofins Genomics). Repair templates were generated by PCR amplification of the Halo or YFP tag flanked by 50 bp homology arms to the target gene, using Q5 High-Fidelity DNA Polymerase (New England BioLabs). PCR products were purified with a PCR purification kit (Blirt, EM26.1). Tachyzoites were mixed with the repair template and 10 µg of the corresponding Cas9 vector, and transfected using the Amaxa 4D-Nucleofector system (Lonza, AAF-1003X). Transfected parasites were allowed to invade

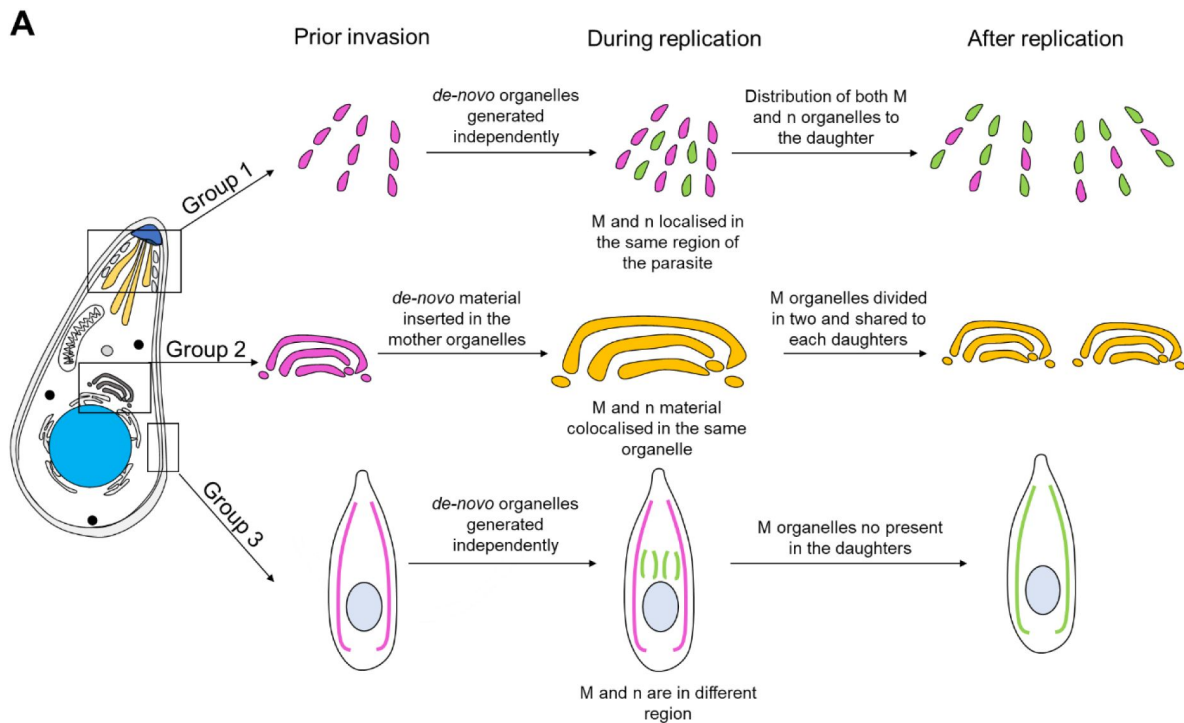


Figure 9

Schematic summary of the three fates of the maternal organelles.

Group 1: Efficient, intact organelle recycling. *De novo* organelles are generated independently of the maternal organelle but in a similar location. Both *de novo* and maternal are distributed to the daughter cells. The Fluorescence intensity of the maternal organelle is relatively stable. Group 2: Even distribution of maternal material through organelle expansion with the insertion of the *de novo* material in the mother organelle. The Fluorescence intensity of the maternal organelle is divided by two at each replication step but their signal surface is increased by two. Group 3: Almost exclusive *de novo* synthesis. The *de novo* organelle is generated independently of the maternal organelle and in a different location. The maternal organelle is not clearly observable after a cycle of replication. The fluorescence intensity of the maternal organelle drastically drops after the first replication. Magenta: maternal organelle, Green: *de novo* organelle, Yellow: Colocalization between maternal and *de novo*.

fresh HFFs and replicate for 48 h. Following manual egress and filtration through a 3 μ m filter, Cas9-YFP-expressing parasites were enriched by FACS (FACSARIA III, BD Biosciences) and sorted into 96-well plates. Correct integration of the repair template was confirmed by PCR.

Labelling and characterization

Maternal and de-novo protein discrimination

Fresh tachyzoites expressing Halo-tagged reporters were mechanically egressed, filtered through a 3 μ m filter, and resuspended in cold DMEM containing a membrane-permeable Halo dye, Janelia Fluor® 646 (1:1000, Promega), for 1 h. Parasites were centrifuged at 2,500 rpm for 5 min and washed three times with fresh medium to remove unbound dye. Parasites were then seeded onto HFF-covered Ibidi live-cell dishes for overnight replication. After replication, a second labeling was performed with a new membrane-permeable Halo dye, Janelia Fluor® 549 (1:1000, Promega) for 1 h at 37 °C, followed by three washes before imaging.

Fluorescence intensity across the replication

Parasites were labeled as described above and allowed to replicate for 24 h on HFF-coated Ibidi live-cell dishes. Approximately 15 fields of view were imaged, and around 25 vacuoles at stages 1, 2, 4, and 8 were analyzed. For each experiment, the maximum fluorescence intensity was measured. Regions overlapping with neighboring parasites or showing non-uniform signal distribution were excluded. Within each biological replicate, the highest recorded value was set to 100%, and all other values were expressed as percentages relative to this maximum. The experiment was performed in three independent biological replicates. Data are presented as mean $\pm$ SD.

Measurement of the signal area of the fluorescence

To quantify the average and total fluorescent area, images were analyzed in Fiji. About 25 isolated vacuoles per stage (1 to 8) were cropped and analyzed individually. A threshold was applied to the maternal signal, and both average and total signal area were measured. Three independent biological replicates were performed, and values are reported as mean $\pm$ SD.

Quantification of percentage of parasite per vacuole with maternal protein

Images were analyzed in Fiji to determine the proportion of parasites per vacuole with maternal protein signal. Approximately 25 vacuoles per stage (1 to 8) were cropped and analyzed individually. Vacuoles where all parasites retained maternal signal were scored as 100%. If some parasites lacked signal, the percentage of maternal-positive parasites was calculated. The experiment was performed in three independent biological replicates. Values are reported as mean $\pm$ SD.

Live replication assay

Parasites were labeled with Janelia Fluor® 646 (1:1000, Promega) for 1 h and washed three times before transfer to Ibidi live-cell dishes covered with HFFs. Parasites were allowed to invade for 1–2 h to obtain ~10 parasites per field of view. Excess parasites were removed by washing three times. Imaging was performed on a Leica DMI8 microscope at 37 °C in 5% CO₂. Images were acquired every 15–30 min for 12 h using minimal laser power and exposure. Experiments were performed in three biological replicates and analyzed in Fiji.

Colocalization assays: ANKER1-Halo with HDEL-GFP

Parasites expressing ANKER1-Halo were transiently transfected with an HDEL-GFP plasmid to label the ER. After overnight replication in HFF-coated Ibidi dishes, parasites were labeled with Janelia Fluor® 646 (1:1000, Promega) for 1 h and washed three times prior to imaging. Z-stacks of HDEL-GFP-positive parasites were acquired, and colocalization was quantified in Fiji. Three biological replicates were performed.

Colocalization assays: MIC2/RON2 with Cb-Emerald

For colocalization with the F-actin network, MIC2 or RON2 was endogenously tagged and labeled in a strain stably expressing Cb-Emerald. Freshly egressed parasites were labeled with Janelia Fluor® 646 (1:1000, Promega) for 1 h, washed, and seeded onto HFF-coated live-cell dishes. After 24 h replication, parasites were fixed in 4% paraformaldehyde and imaged. Z-stacks of vacuoles showing clear F-actin networks were acquired, and colocalization was analyzed in the cytoplasm and residual body. Three biological replicates were performed. Data are reported as mean $\pm$ SD. Images were analyzed in Fiji.

Quantification of the maternal microneme distribution

To assess the distribution of maternal micronemes at stage 8, parasites were labeled as described in “Maternal and de novo protein discrimination.” Twenty-five stage 8 vacuoles were selected. Using the de novo MIC2 signal, the outline of each tachyzoite was traced, and the number of maternal MIC2-positive parasites was counted. Tachyzoites were numbered 1 to 8 from left to right. Three biological replicates were performed. Data are reported as mean $\pm$ SD. Images were analyzed in Fiji.

Phenotypic assays

Induction of MyoF KD

MyoF-mAID parasites were genetically modified to endogenously tag MIC2, RON2 and SortLR. As previously described for MyoF mAID parasites (Carmeille et al., 2021 [DOI](#)), MyoF-mAID MIC2/RON2 or SortLR-Halo parasites were induced +/- auxin for 4h prior to labelling and experiments.

Quantification of the accumulation of protein in the residual body

Parasites were treated $\pm$ auxin for 4 h and labeled as described above. After 24 h replication on live-cell dishes, ~100 vacuoles were imaged across ~15 fields of view. The percentage of vacuoles showing accumulation of material in the residual body was calculated. The experiment was performed with three independent biological triplicates. The values are then expressed as the mean values of the three independent experiments $\pm$ SD. Images were analysed via Fiji.

Auxin chased experiment

Parasites were induced +/- auxin for 4h prior labelling as describe above in “Maternal and de-novo protein discrimination”. After transfer on live cell dish and 48 h of replication +/- auxin, the parasites were images. For the chased condition, the parasites were grown for 24h under auxin. After this initial 24h, the dishes were washed three time to completely remove the auxin, a fresh media (without auxin) was added. Parasites were then allowed to replication for an extra 24h. About 15 fields of view were images for a total of about 100 vacuoles. The percentage of vacuole with accumulation in the residual body was calculated. Some parasites were exhibiting a redistribution where the microneme were no longer stuck in the residual body but not as homogeneously distributed as control parasite microneme. This profile was name “redistributed”

and quantified along with the “accumulated” and “normal” patterns. The experiment was performed with three independent biological triplicates. The values are then expressed as the mean values of the three independent experiments $\pm$ SD. Images were analysed via Fiji.

MyoF-mAID Live replication assay

Parasites were induced +/- auxin for 4h prior labelling. Parasites were labelled with the Halo membrane permeable dye Halo 646 (Janelia Fluor®646 1:1000 Promega) for 1 h and washed three-time prior transfer onto a live cell dish covered with HFF. Parasites were allowed to invade for 1h-2h to obtain enough invasion events to reach an average of 10 parasites per FOV. After the invasion, the excess of parasites was removed by 3 washes. A new membrane permeable dye Halo 549 (Janelia Fluor®549 1:10000 Promega) was added to the media to record the de-novo generation of the POI live. Live cell dishes were then transferred to the Leica-DMI8, heated at 37°C under a chamber containing 5% CO₂. Laser power and exposure were adjusted to the lowest values allowing reliable imaging. Images were taken every 15-30 minutes to follow the tagged protein behaviour during replication for 12h. For the Chased assay, the imaging started after the removal of the auxin after 24h of replication as described above in “Auxin chased experiment.” The experiments were performed with three independent biological triplicates. Images were analysed via Fiji.

Imaging

Widefield microscopy

Unless stated otherwise, all images were acquired on a Leica-DMI8, objective 100x with the LasX software (v3.7.4). Fiji (v1.53c) was used to analyse the picture and all counts were made manually. LasX software (v.) from Leica was used to obtain parasite imaging data and

All images and movies were processed using Fiji (ImageJ) software v Image Processing Software (Schindelin et al., 2012 [link](#)).

Software

Fiji (FIJI ImageJ v1.54f) was used to analyze the picture and all counts were made manually.

Data analysis

All data were plotted using Microsoft Excel

Statistical analysis

For statistic relevance, one-tailed unpaired Student's T-tests were performed.

Supplementary Figures

Figure S1

ANKER1 is a resident of the ER.

A) Representative picture of the colocalization ANKER1-Halo with HDEL-GFP transfected parasites. Magenta: ANKER1-Halo, Green: HDEL-GFP. **B)** Quantification of the colocalization between ANKER1-Halo and HDEL-GFP. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.

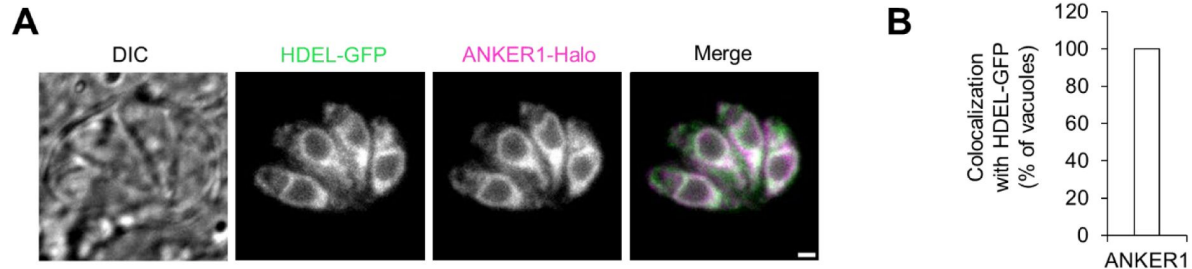
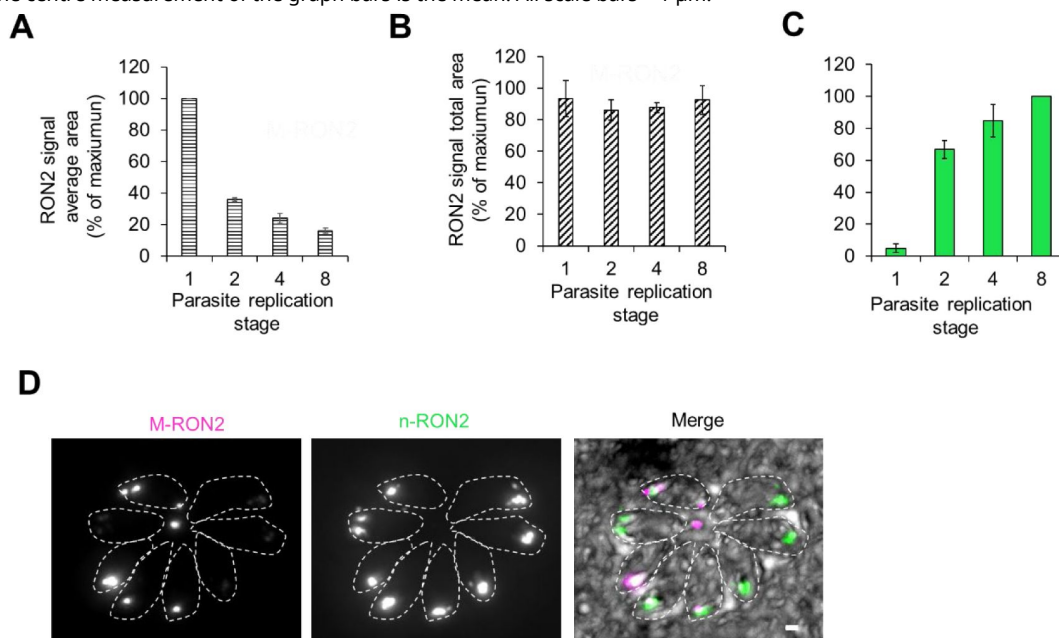


Figure S2

Rhoptries are inherited intact.

A) Quantification of the average signal area of M-RON2 during replication. The average surface of M-RON2 decrease at each replication suggesting a separation of the mother organelle. **B)** Quantification of the total signal area of M-RON2 during replication. Despite the decrease of the average signal area, the total surface of M-RON2 remain stable indicating that the totality of the maternal organelles are conserved during replication. **C)** Fluorescence intensity analysis of n-RON2. **D)** Representative picture of a vacuole with missing maternal rhoptries in some daughter cell of a single vacuole at stage 8. Magenta: M-RON2, Green: n-RON2. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.



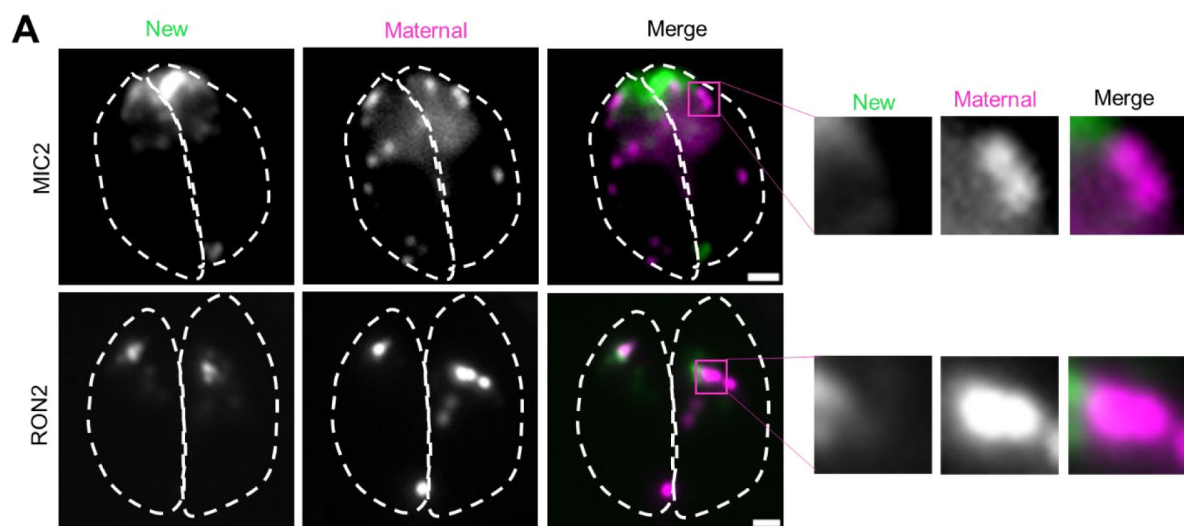


Figure S3

The *de novo* secretory organelles are generated independently of the maternal.

A) The *de-novo* secretory organelle are generated independently of the maternal organelles. Representative picture of MIC2 and RON2-Halo parasites. Magenta: Maternal, Green: *de novo*. Zoom window highlight the absence of colocalization between maternal and *de novo* material.

Three biological replicates were used. All scale bars = 1 μm .

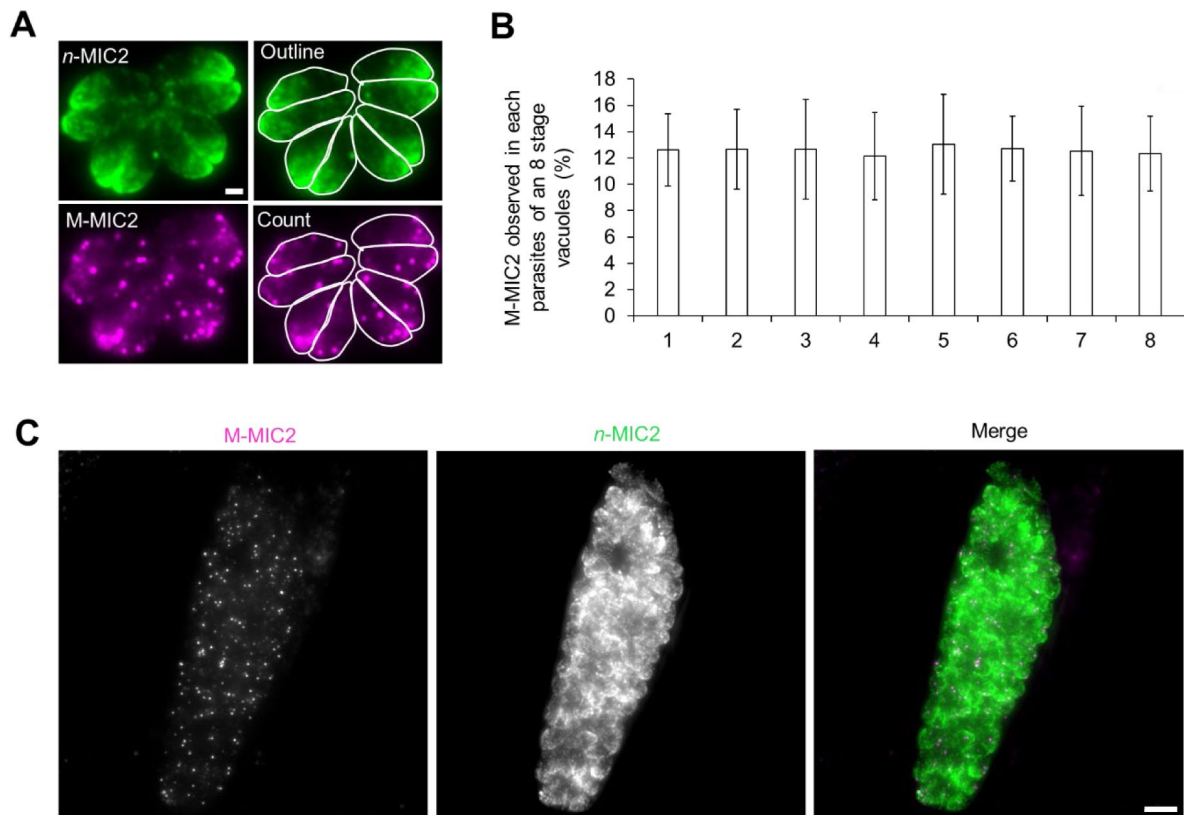


Figure S4

Maternal micronemes are evenly distributed to daughter cells.

A) MIC2-Halo parasites at stage 8 were stained for maternal (M-MIC2, magenta) and de novo (n-MIC2, green) MIC2. n-MIC2 signal was used to outline each tachyzoite, numbered 1–8 left to right, and M-MIC2 micronemes were counted per cell. Scale bar = 1 μ m. B) Quantification of average M-MIC2 micronemes per tachyzoite shows consistent numbers across daughters, indicating equal distribution. C) Representative image of a late-stage vacuole shows uniform M-MIC2 distribution among all daughter cells. Scale bar = 5 μ m. Three biological replicates were used; bars show means $\pm$ SD.

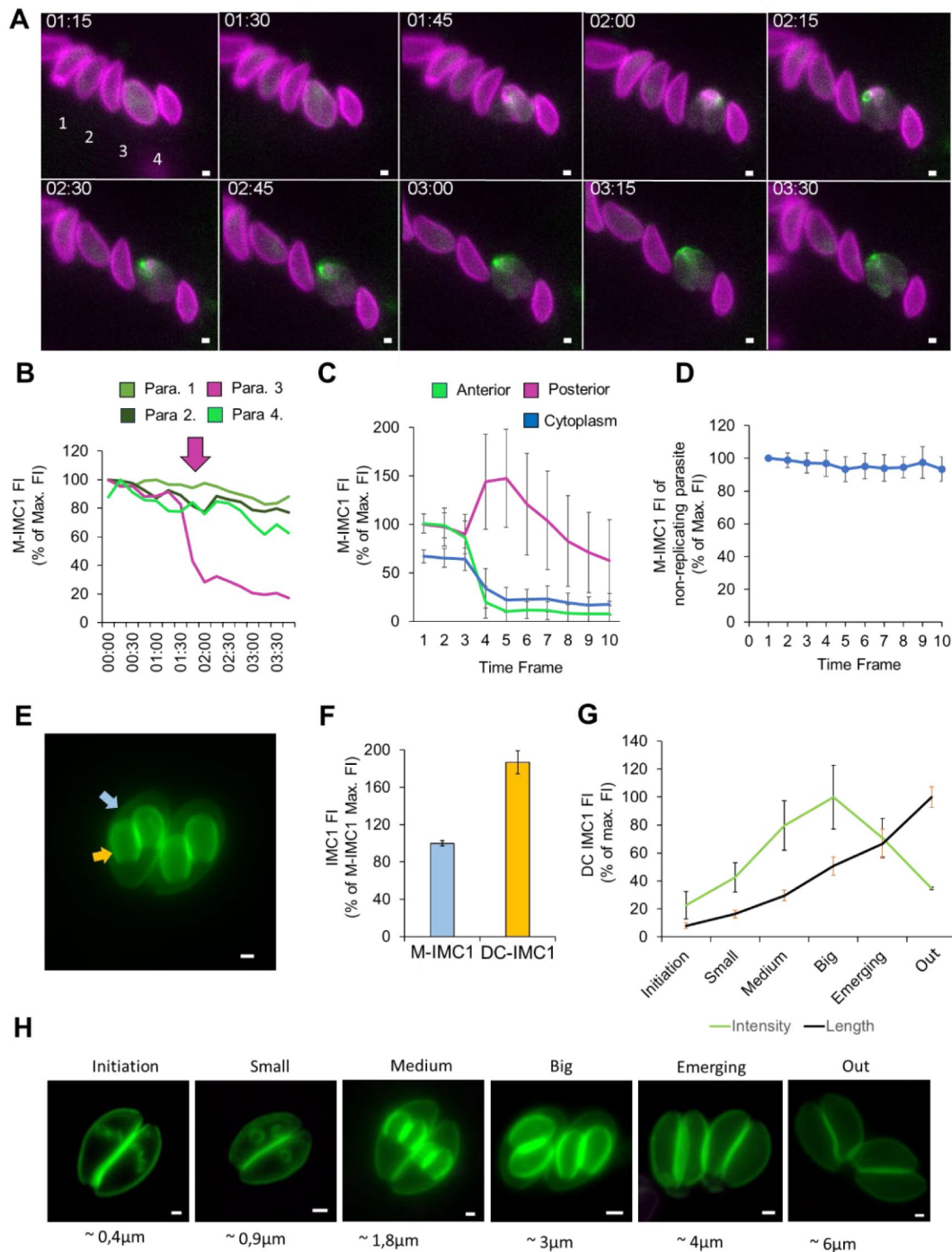


Figure S5

IMC degradation occurs after daughter cell formation but before budding.

A) Time-lapse of IMC1-Halo during the first replication cycle, with F-actin visualized via chromobody-emerald. Magenta: M-IMC1, Green: F-actin. M-IMC1 accumulates in the residual body with no redistribution or signal loss in non-replicating parasites. B) Fluorescence intensity tracking of parasites 1–4 (see A) during replication of parasite 3. C) Mean fluorescence intensity across apical, cytoplasmic, and basal regions in 25 replicating parasites. D) Mean M-IMC1 intensity in 25 non-replicating parasites during the same timeframe as C. E) Image showing daughter cells forming within the mother. Green: IMC1-YFP. Blue arrow: mother; yellow arrows: daughters. F) IMC fluorescence comparison between mother and daughters, with mother set to 100%. G) IMC intensity in daughters relative to size. Peak intensity occurs when daughters reach $\sim 3 \mu\text{m}$, just before emergence. H) Representative images of daughter cells at different sizes used for classification. Three biological replicates were used; bars show means $\pm$ SD. Scale bars = $1 \mu\text{m}$.

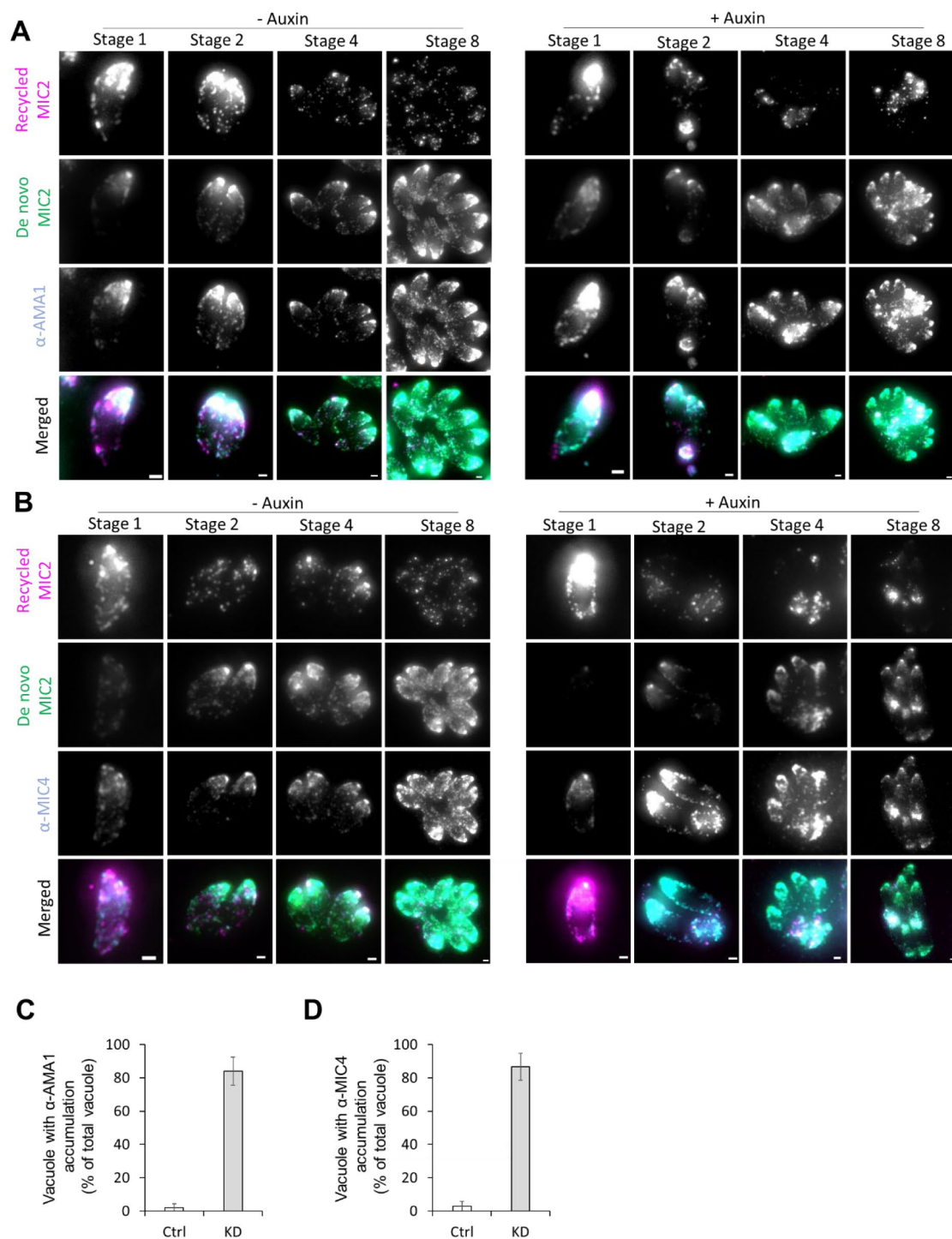


Figure S6

AMA1 and MIC4 recycling is impaired in the absence of MyoF.

A) Effect of MyoF knockdown (KD) on AMA1 inheritance. MyoF-mAID MIC2-Halo parasites were grown $\pm$ auxin for 24h. Magenta: M-MIC2, Green: n-MIC2, Cyan: α -AMA1. Without MyoF, AMA1 accumulates in the residual body, mirroring the MIC2-Halo pattern. B) Effect of MyoF KD on MIC4 inheritance. Magenta: M-MIC2, Green: n-MIC2, Cyan: α -MIC4. MIC4 also accumulates in the residual body over time in the absence of MyoF. C-D) Quantification of vacuoles showing α -AMA1 (C) and α -MIC4 (D) accumulation. White: control; Gray: MyoF-KD. Three biological replicates were used; bars show means $\pm$ SD. Scale bars = 1 μ m.

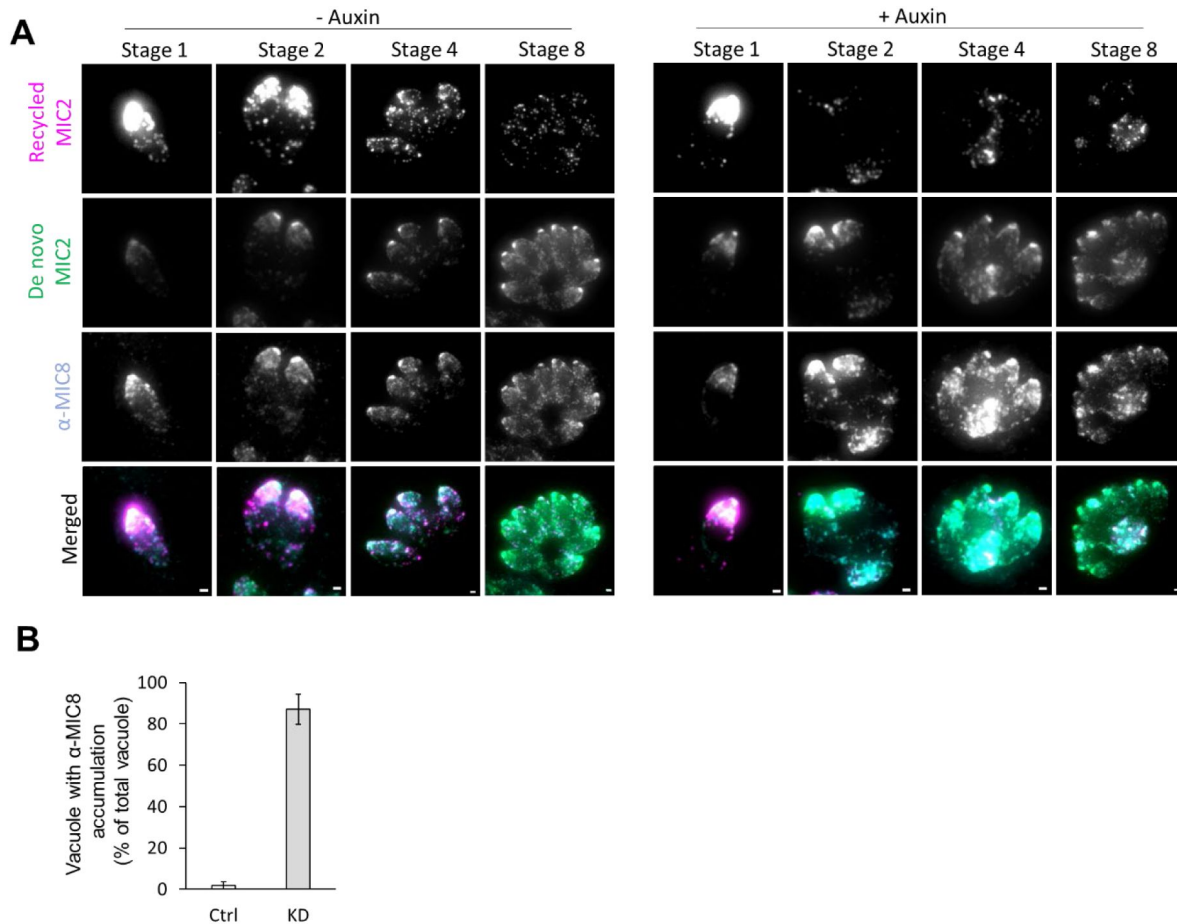


Figure S7

In absence of MyoF, MIC8 is also blocked in its recycling.

A) Impact of MyoF KD on MIC8 inheritance during replication. MyoF-mAID MIC2-Halo parasites were labelled and grown for 24h +/- auxin. MIC8 was visualised using antibodies. Magenta: M-MIC2, Green: n-MIC2, Cyan: α -MIC8. In absence of MyoF, as observed for all the other microneme markers, α -MIC8 accumulate as replication goes on, following a similar pattern as MIC2-Halo. **C)** Quantification of the percentage of vacuoles exhibiting accumulation of α -MIC8 in the residual body. White: Control, Gray: MyoF-KD. Three biological replicates were used for all analyses; error bars are standard deviations, and the centre measurement of the graph bars is the mean. All scale bars = 1 μ m.

Acknowledgements

This work was supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) grant GR 5696/2-1 to Simon Gras, grant ME 2675/6-2 and DFG Equipment grant INST 86/1831-1 to Markus Meissner. We thank VEuPathDB for their invaluable Informatics Resources. The MyoF-mAID strain was kindly provided by Prof. Aoife Heaslip (University of Connecticut), MIC4, MIC8 and AMA1 antibodies were kindly provided by Prof Dominique Soldati-Favre (University of Geneva) and Prof. Gary Ward (University of Vermont) respectively. We thank Dr. Elena Jimenez-Ruiz for the useful discussions.

Additional files

Video S1. Live degradation of the maternal IMC1. Magenta: M-IMC1 (Jan646), Green: Cb-Emerald. [↗](#)

Video S2. Live inheritance of the maternal RON2. Magenta: M-RON2 (Jan646), Green: IMC1-YFP. [↗](#)

Video S3. Live inheritance of the maternal MIC2. Magenta: M-MIC2 (Jan646), Green: IMC1-YFP. [↗](#)

Video S4. Live replication of MyoF mAID MIC2-Halo without auxin. Magenta: M-MIC2 (Jan646), Green: n-MIC2 (Jan549). [↗](#)

Video S5. Live replication of MyoF mAID MIC2-Halo with auxin. Magenta: M-MIC2 (Jan646), Green: n-MIC2 (Jan549). [↗](#)

Video S6. Live replication of MyoF mAID MIC2-Halo after auxin chase. Magenta: M-MIC2 (Jan646), Green: n-MIC2 (Jan549). [↗](#)

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Reviewer #1 (Public review):

Summary:

This work asks the question of how different organelles and structures in the apicomplexan parasite *Toxoplasma gondii* are recycled and/or segregated to the daughter cells during cell replication. In particular, they consider an unusual cell structure called the residual body that links replicating cells during the intracellular infection stage of this parasite. The residual body has historically been considered a 'dumping ground' for unnecessary relics of the mother cell during division, but this notion is increasingly being revised. Indeed, cell replication in *Toxoplasma* is often misinterpreted as cell division (cytokinesis), but in fact, the cell replicates its organelles and structures to multiple 10s of copies in seemingly distinctly formed daughter cells, but cytokinesis is delayed for many such cycles and typically only occurs simultaneously with parasite egress from its host cell. The residual body is, in fact, the

connection between these pre-cytokinetic replicated daughters, and effectively, this is still a single cell at this stage. The authors have previously shown that an actin network extends through the residual body between these daughter cells, and ER and mitochondria common to all cells are also linked through this structure. This study examining the fates of organelles during cell replication is timely for continuing our understanding of how this fascinating component of the cell participates in these processes. The authors use Halo-tags as their principal tool to track discrete populations of proteins, labelling their organelle locations, and this provides beautiful insight into these processes.

Strengths:

Using dyes conjugated to Halo tags, this work elegantly tracks the fates of proteins synthesised by an original 'mother' cell over several replication cycles of pre-cytokinetic 'daughters'. Using this tool, they show that some organelles are made intact just once and that some of these can be subsequently sorted to the daughters (micronemes and rhoptries) while others are dismantled (IMC) and the daughters must make their own. A third set of organelles (largely synthesis, sorting, and metabolic compartments) is divided and inherited, and new daughter-synthesised proteins are added to the preexisting maternal proteins in these structures. A role for actin and myosin is clearly demonstrated for micronemes and rhoptries, and this correlates with their relatively late inheritance into the developing daughters. Overall, this work gives clarity to the behaviours of several cell structures during replication and paves the way to a better understanding of the mechanisms that drive the differences between structures and the universality of these processes in other apicomplexan parasites.

Weaknesses:

In addressing the question of residual body participation in sorting of organelles, it would be useful to clearly define this structure and when and where it is delineated from the posterior of a mother cell during the formation of daughter structures. This might seem like a moot point, but it would give clarity to notions of recycling and 'reservoirs'. Mother cells retain their active invasion apparatus until very late in daughter formation, and the need for micronemes and rhoptries to be released from this service late in the process might explain why they are only then trafficked to the cell posterior and then into the daughters. So, is this a distinct 'residual body' body function/reservoir or just a spatial constraint of this sequence of daughter formation? In subsequent cell replications (4, 8, 16... stages), is there a separation between the residual body that links them all and the posterior of each new 'mother cell', and if so, when is this distinction lost? This is important because without a definition, we might be confusing different processes. Are rhoptries/micronemes that originate in one 'mother' able to be sorted to the 'daughters' from a distinct mother in this syncytium? If so, this would make it a sorting centre, but otherwise we could be just capturing the activities at the posterior of any given cell during replication. The authors' further thoughts on this would be very interesting.

The Group 2 structures are described as those that are divided between daughters and receive newly synthesised proteins that add to the maternal protein of these compartments. While this is a logical conclusion for several that are mentioned, where the maternal protein signal is seen to be depleted with replication (including for the apicoplast, ER, glideosome, and Golgi). Data for the addition of new proteins to these existing structures is actually only presented in direct support of this for the Golgi.

<https://doi.org/10.7554/eLife.108685.1.sa2>

Reviewer #2 (Public review):**Summary:**

Toxoplasma gondii is an obligate intracellular parasite and the causative agent of Toxoplasmosis. Parasite invasion into host cells, intracellular replication, and then egress, which results in the destruction of the infected cell, is central to pathogenicity. This manuscript focuses on understanding how maternal resources (in this case, cellular organelles) are shared between daughter parasites during cell division. Many organelles are single copy, meaning that division and inheritance by the daughters is crucial for successful replication. The major strength of this study was the use of a Halo-based pulse chase assay to characterize patterns of organelle inheritance. The results show that both microneme and rhoptries (secretory vesicles) previously thought to be synthesized *de novo* are inherited by daughter parasites. Thus, this paper adds new insight to our understanding of cell division in this important parasite.

Strengths:

This study demonstrated that pulse labeling of proteins can be used to monitor protein synthesis, turnover, and movement. This approach will be of great interest to the field. Using this method, the authors demonstrate three main modes of organelle inheritance.

(1) Organelles, where there are multiple copies (such as secretory vesicles, micronemes, and rhoptries), are divided between the daughter parasites, with additional contribution of newly formed vesicles. New and old material remain as separate entities in the cell.

(2) Single-copy organelles, which are expanded to include newly synthesized material prior to division, such as the Golgi and apicoplast.

(3) Cytoskeletal structures that are synthesized anew during each round of division. These studies provide more refined insight into patterns of organelle inheritance and demonstrate that secretory organelles are not made *de novo* during each round of division as previously thought. The paper has a logical flow, and overall, the data is presented in a clear and organized fashion.

Weaknesses:

(1) Descriptions of methodology and statistical analysis were incomplete.

(2) There are inconsistencies between the data in Figures 1 and 5. In Figure 1, a small amount of maternal IMC is visible in stage 2 parasites. Although this is a ~90% reduction, these parasites should be quantified as parasites with maternal IMC. However, the graph in Figure 5C indicates that no maternal parasites have GAPM1a, given that graph 5C is a binary measure (present vs. absent), one would expect a non-zero percent of parasites to have maternal material.

(3) The conclusion from Figure 6 was not justified based on the data. I agree with the author's conclusion that the accumulation of micronemes and rhoptries in the residual body was time-dependent. In Figure 6A, the signal observed in the residual body at times 6:30, 13, and 14 hours is not observed in subsequent time points. However, the fate of these micronemes and rhoptries is unclear. It cannot be concluded that these vesicles are recycled back to the mother. They could also have been degraded. In fact, the graphs of microneme inheritance in Figure 2B show a decrease in maternal signal from 100% to 80% between stages 1 and 2, indicating that some microneme degradation is taking place.

(4) To convincingly demonstrate that the redistribution of micronemes and rhoptries was due to recovery of MyoF protein levels after auxin washout, a Western blot should be performed

to show MyoF protein levels over time. In addition, the decrease in mMIC2 protein levels in the residual body in Figure 8F should be measured and normalized for photobleaching. Both apical and basal signals appear to be reduced over the time course of imaging.

<https://doi.org/10.7554/eLife.108685.1.sa1>

Reviewer #3 (Public review):

Summary:

Knoerzer-Suckow et al. explore the mechanisms of organelle inheritance during endodyogeny in *Toxoplasma gondii* using an innovative dual-labeling approach to track the distribution of maternal organelles into daughter parasites. They can clearly distinguish between maternal and daughter-derived organelles using their dual-labeling Halo Tag approach. They reveal that different organelles are trafficked to daughter parasites in three broad patterns, which they have binned into groups. Their findings reveal a role for MyoF in the inheritance of micronemes and rhoptries, and notably, they observe that the inner membrane complex (IMC) is not recycled. Instead, the IMC undergoes a pronounced relocalization to the posterior of the maternal cell, where it is likely targeted for degradation.

Strengths:

The data surrounding their MyoF knockdown experiments, IMC degradation, and trafficking of MIC2 after auxin washout are compelling. These data add to the knowledge of how organelle inheritance occurs in *T. gondii*, increasing the field's understanding of endodyogeny.

Weaknesses:

- (1) The evidence provided to support the claim that microneme and rhoptry inheritance specifically traffics through the residual body does not sufficiently substantiate the claim. The temporal resolution of the imaging is inadequate to precisely trace the path of microneme and rhoptry inheritance. From the data shown in the manuscript, it can be concluded that at least some of the micronemes and rhoptries might be recycled through the residual body, but it is unclear whether many or most of these organelles do so.
- (2) The absence of specific markers for the residual body brings into question whether microneme inheritance occurs through a discrete residual body or simply via the basal end of the maternal parasite. The authors need a robust way to visualize and define the residual body to claim that micronemes and rhoptries are specifically transported through this structure.

<https://doi.org/10.7554/eLife.108685.1.sa0>